Big Data Spark Hands-On 1

spark-shell

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Objectives

1/ Install on your (Windows) PC a minimalist local SPARK

2/ Configure it, launch spark-shell

3/ Discover spark-shell scala> REPL

4/ Exercises 1,2,..16 : … to Execute basic spark commands on DataSets

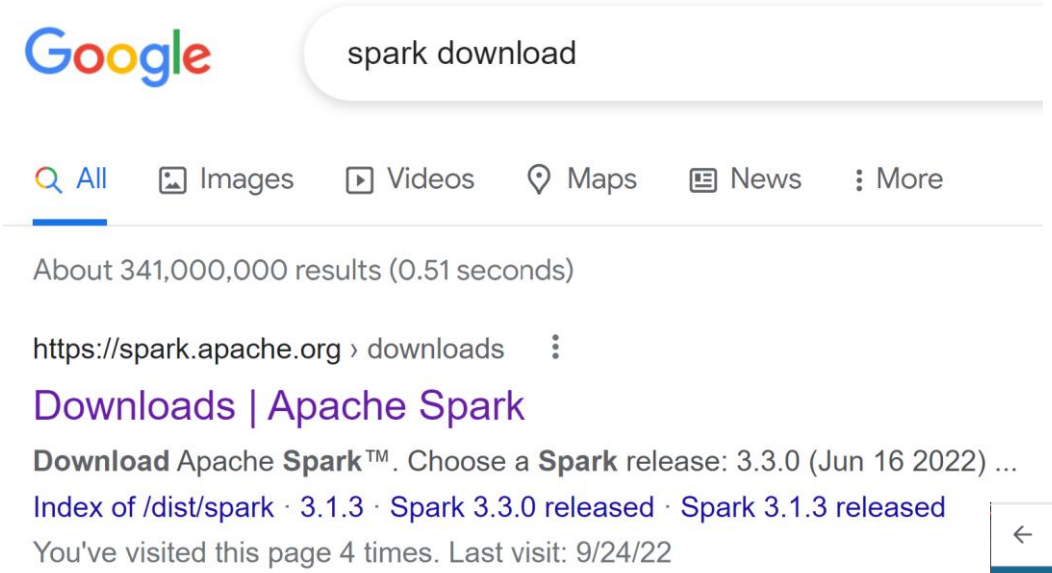
Installing (NOT on Windows..) with a Package Manager : MacOS, Linux

MAC OS:
brew install apache-spark

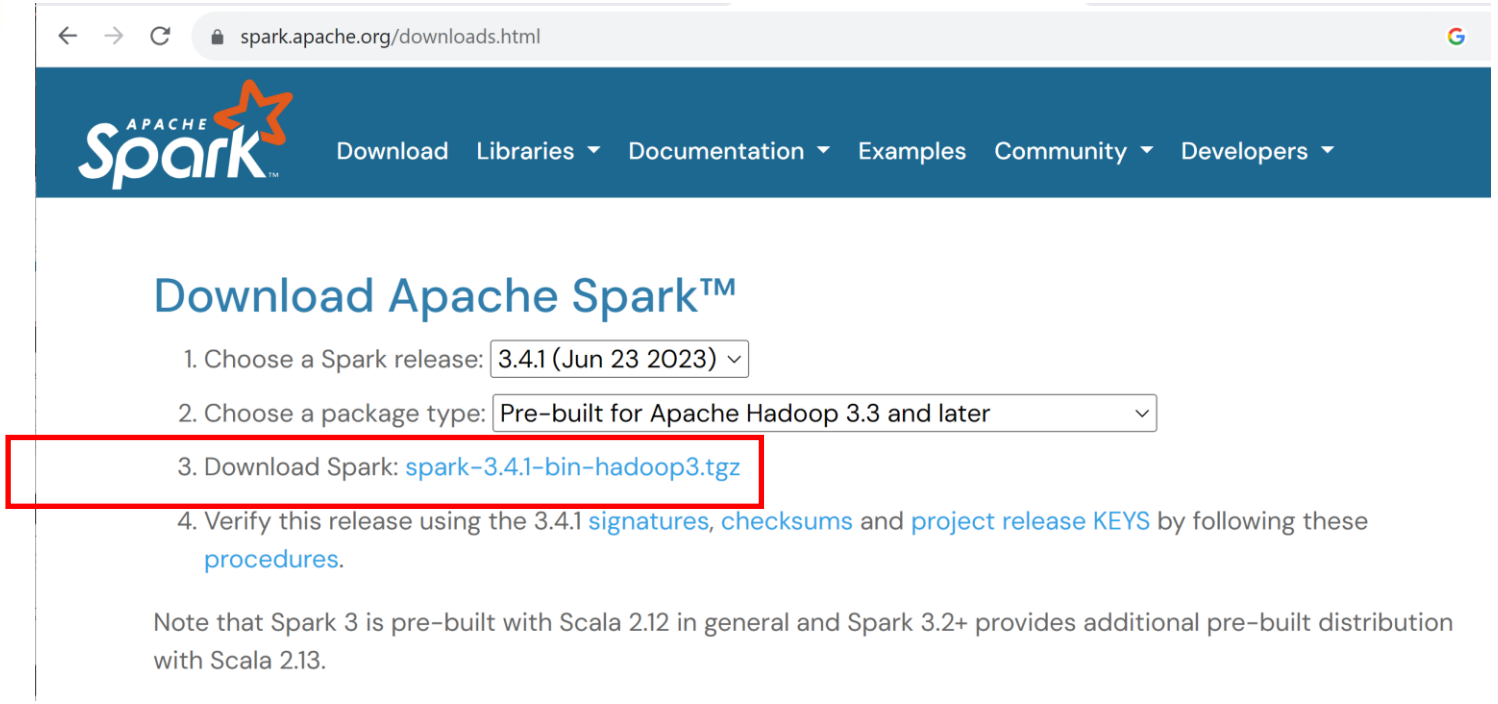
RedHat:
yum install spark

Debian:
sudo apt install openjdk-8-jdk
sudo apt-get install spark
or … sudo snap install spark --classic

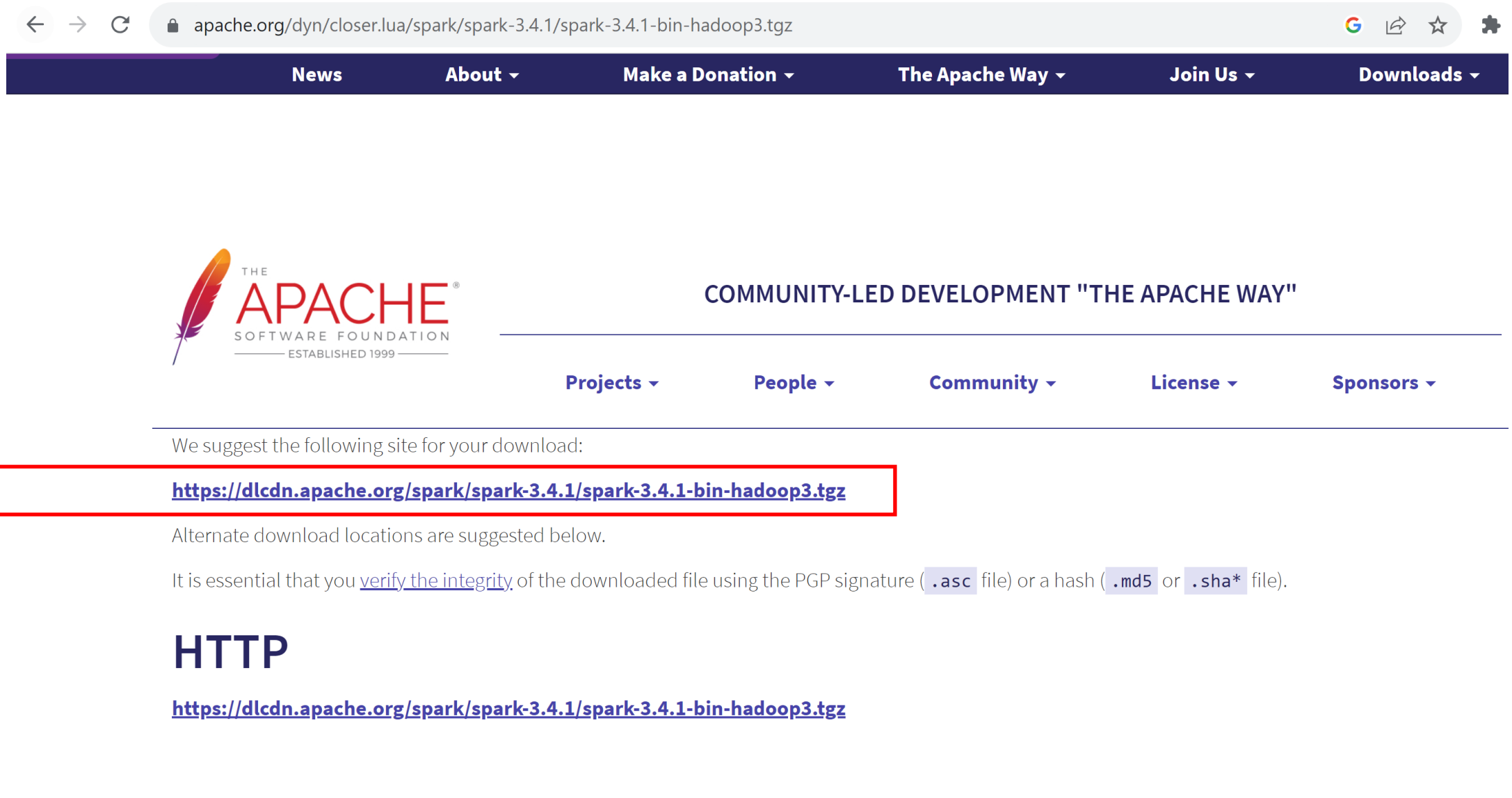
Step 1: Download Spark



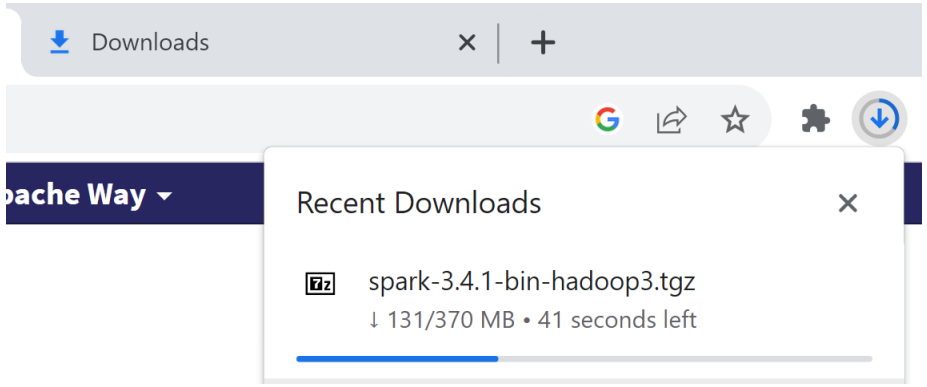
https://spark.apache.org/downloads.html



Download..


apache.org/dyn/closer.lua/spark/spark-3.4.1/spark-3.4.1-bin-hadoop3.tgz
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THE APACHE SOFTWARE FOUNDATION
ESTABLISHED 1999
COMMUNITY-LED DEVELOPMENT "THE APACHE WAY"
Projects
People
Community
License
Sponsors
We suggest the following site for your download:
https://dlcdn.apache.org/spark/spark-3.4.1/spark-3.4.1-bin-hadoop3.tgz
Alternate download locations are suggested below.
It is essential that you verify the integrity of the downloaded file using the PGP signature (.asc file) or a hash (.md5 or .sha* file).
HTTP
https://dlcdn.apache.org/spark/spark-3.4.1/spark-3.4.1-bin-hadoop3.tgz

Download ..
spark3=370 MB, spark4=530 MB

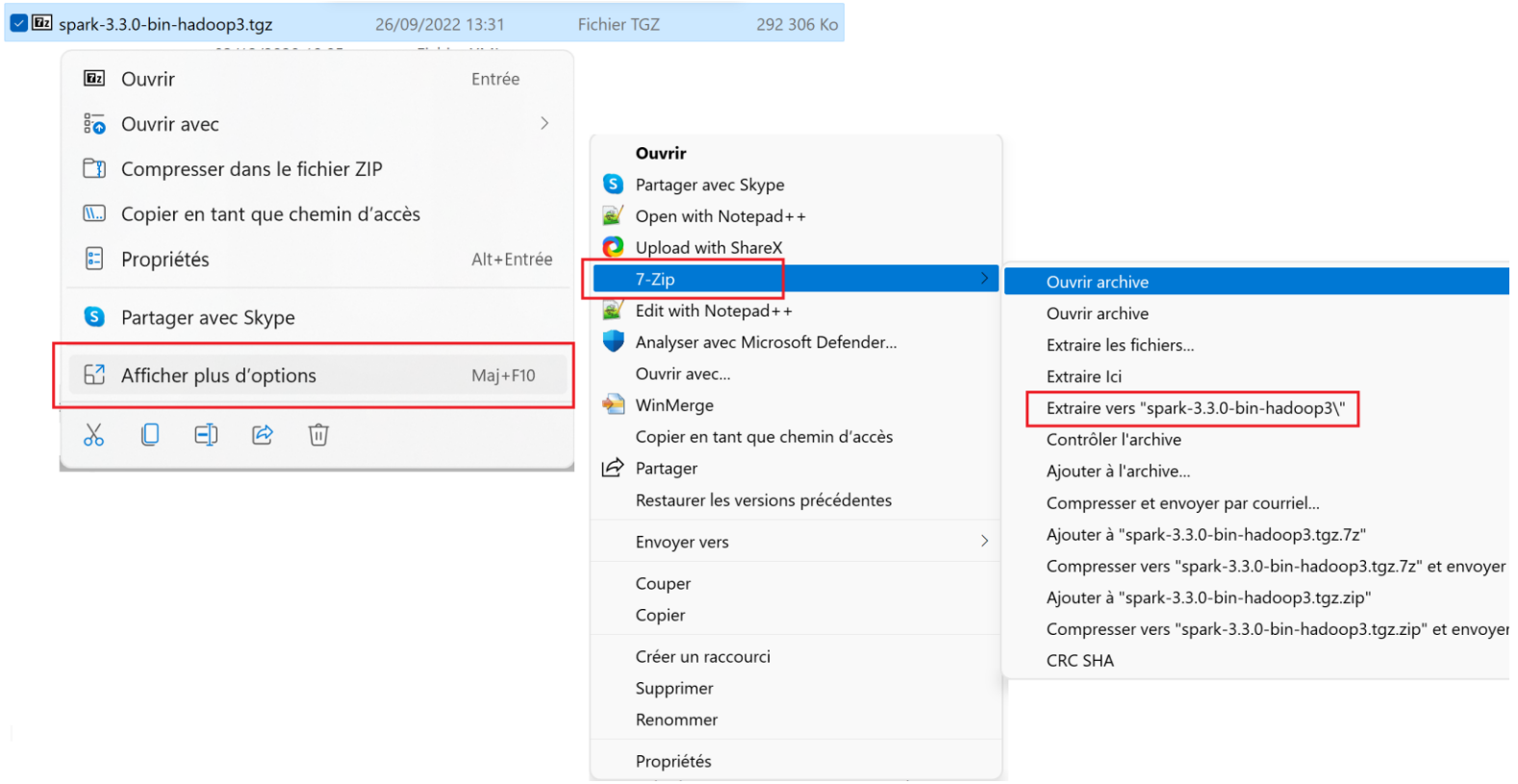


(cf also Step 3 : another download for JDK next !! … 100MB more)

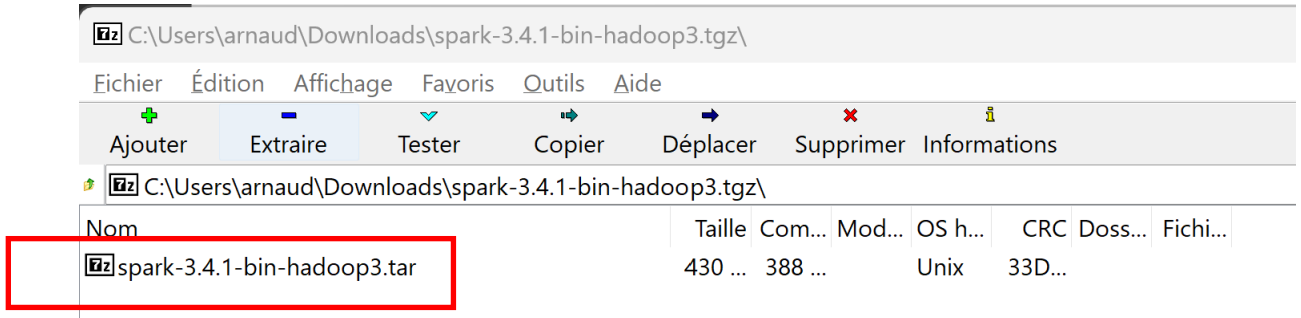
Step 2: Unzip
for example in C:\apps\spark

From cygwin / Linux ? => simply type « tar zxf spark*.tgz »

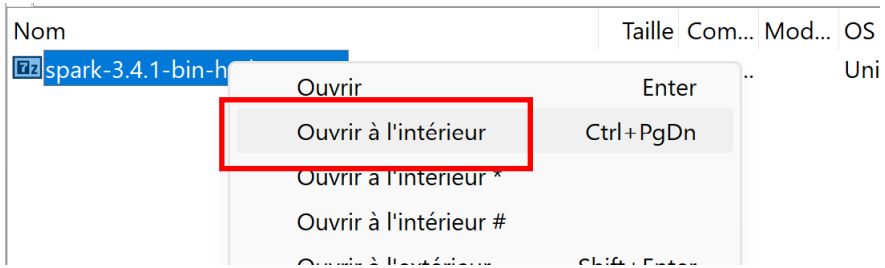
From Windows …

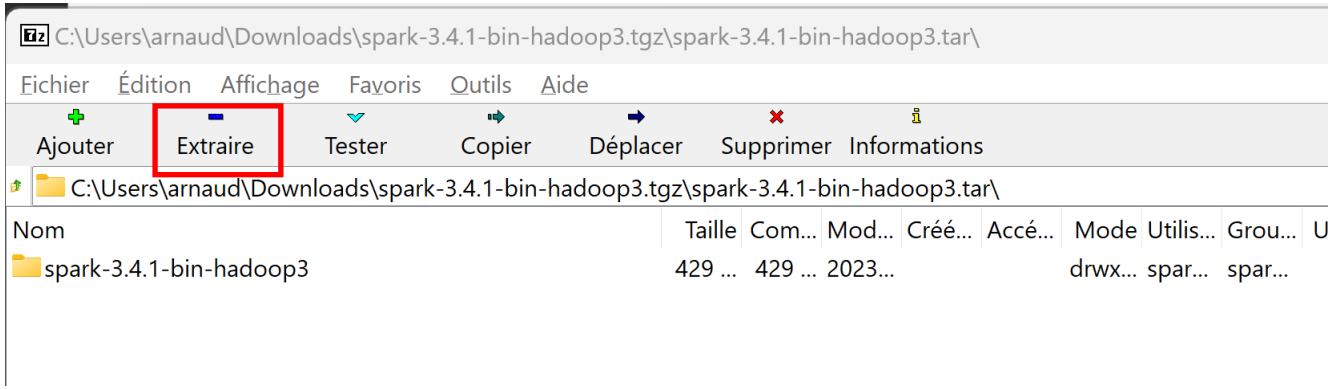


7-Zip … not very user-friendly for « .tar.gz »

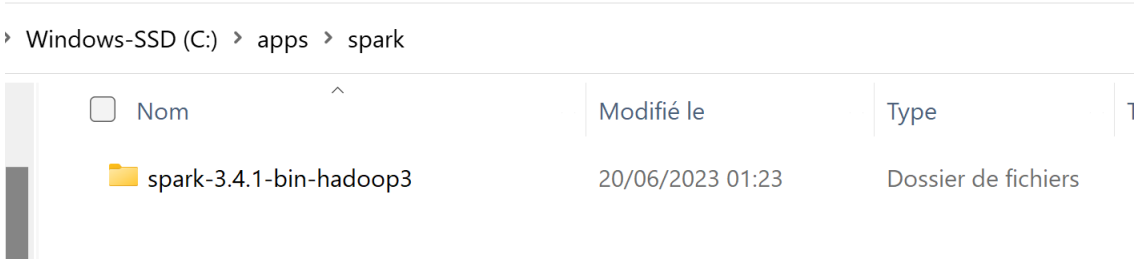


First unzip …. 7-Zip only understand the « .gz » suffix out of « .tar.gz »
=> You extract a « .tar » file …. Then you redo extract from this file !



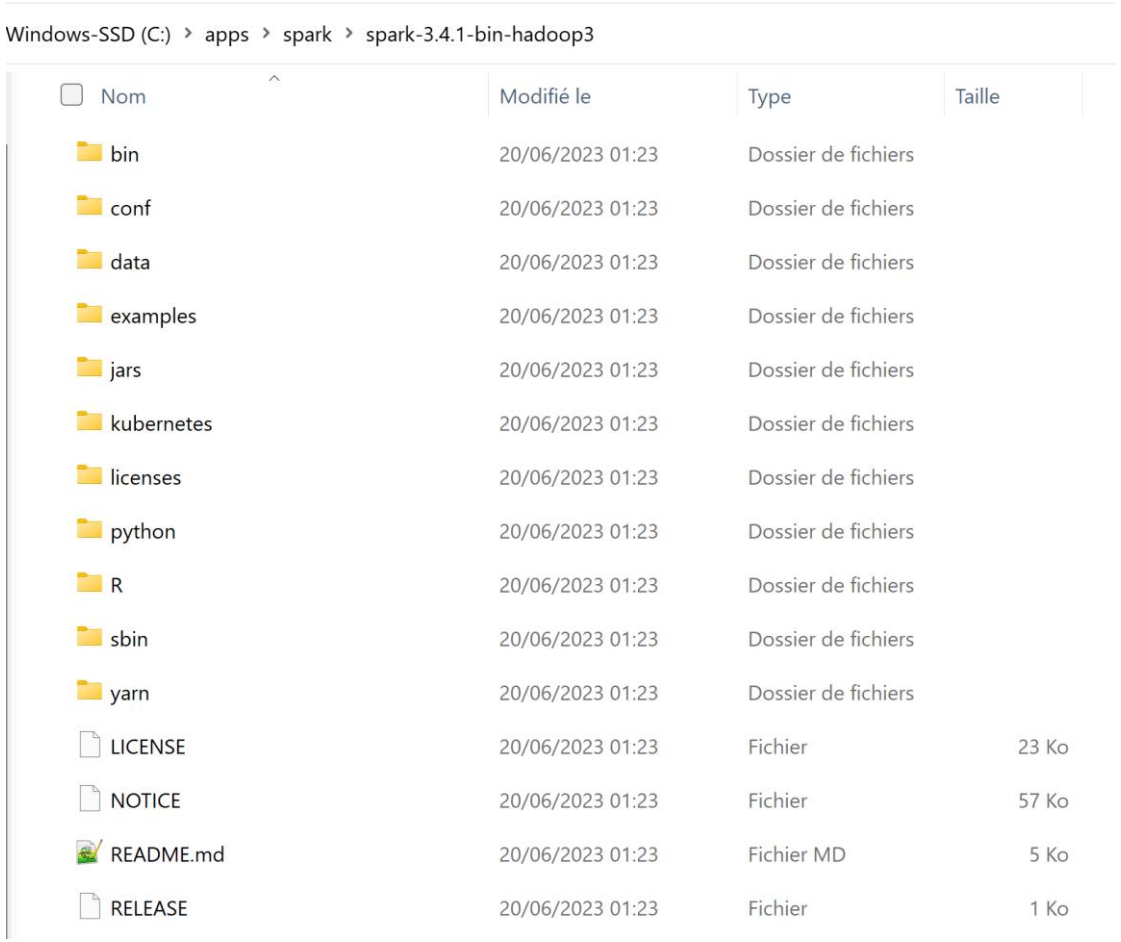


Step 2… check extract « .tar » file!



Simplify… move and rename sub-sub dir to « c:\apps\spark »

your version may be 3.4.xx
!= 3.3.0 … screenshot from oct 2022
warn on next slides for your exact dir path!



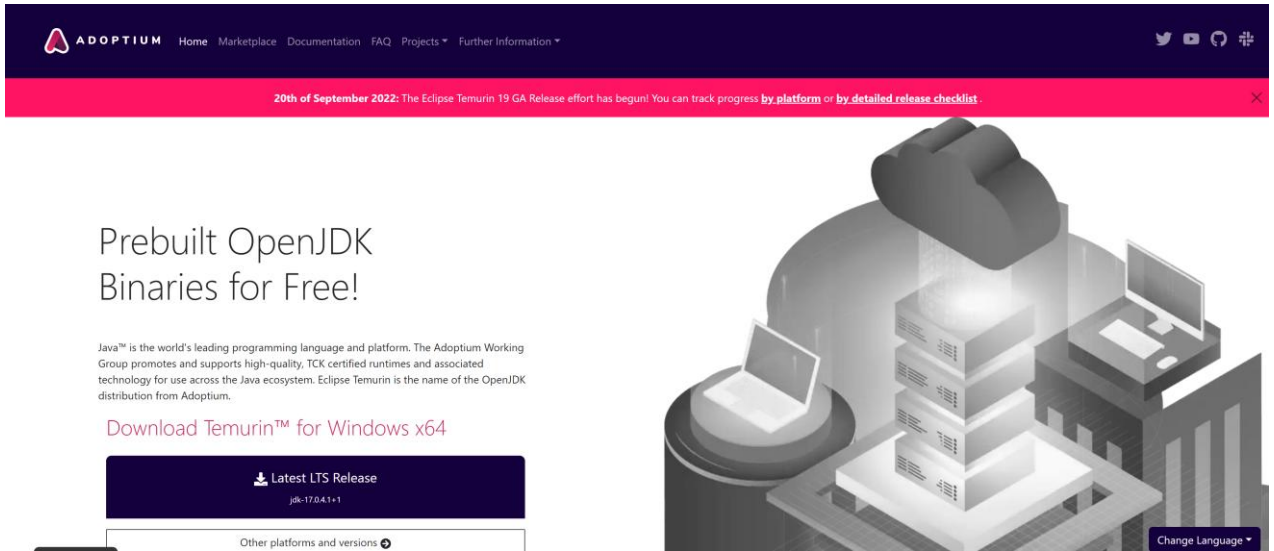
Step 3: Pre-requisite
Install JDK

Notice: most of Hadoop still use Java 8+linux.. But cmd error on windows?

JDK is Open-Source, but pre-built binaries may be licenced if downloading from oracle.com

Download from « adoptium » (previously « adoptOpenJdk »)

https://adoptium.net/



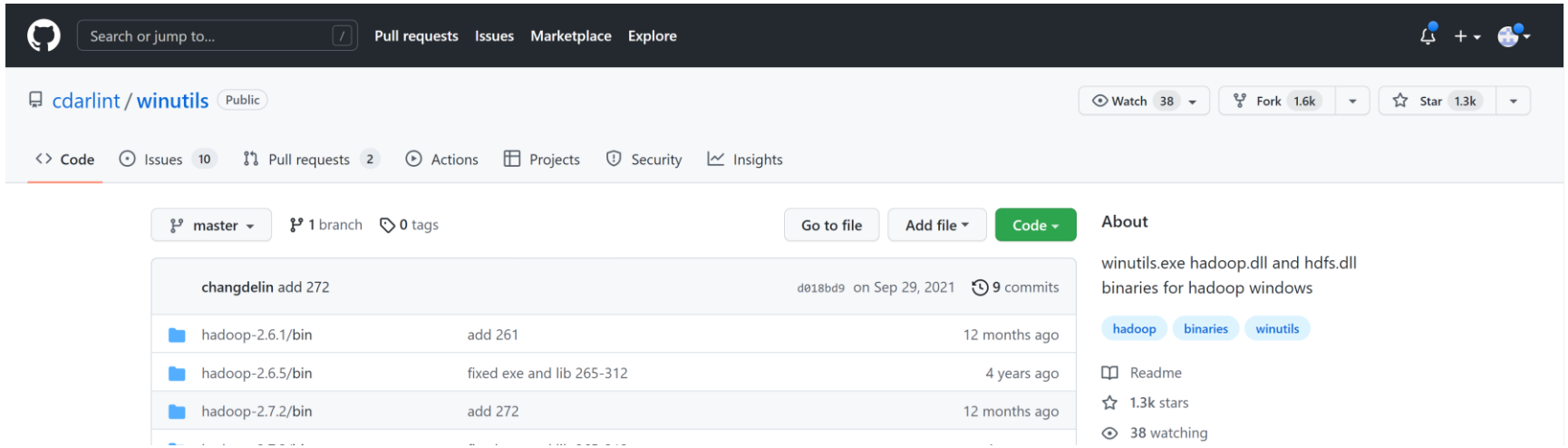
Step 4: on Windows only
Download « WinUtils »

copying 2 files => **from DVO > Course Spark > « TD1 »**
hadoop.dll + winutils.exe in your Spark\bin

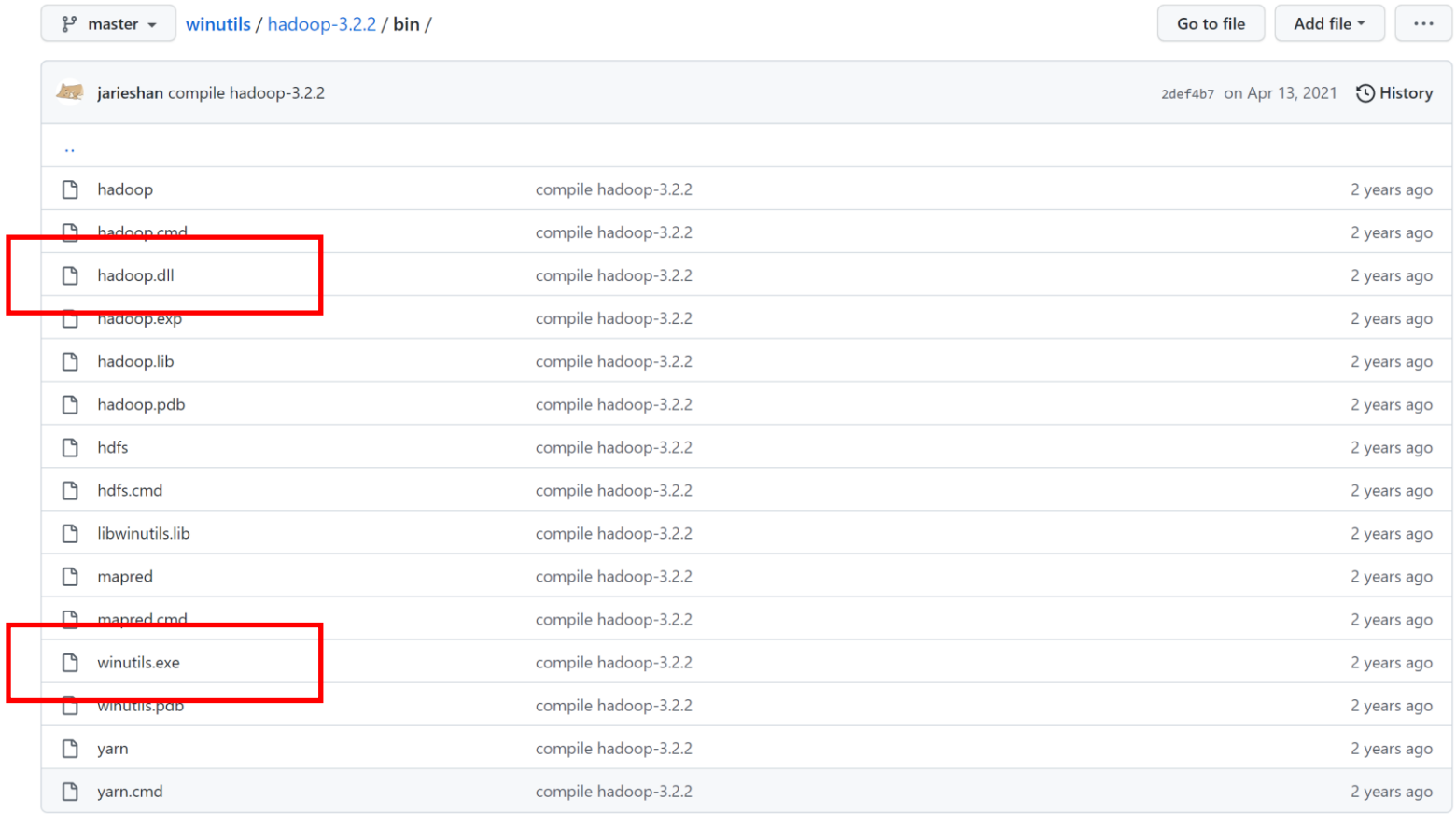
OR ELSE (more difficult)
If you want to copy
from Source directly

.. Cf 2 Files to extract from

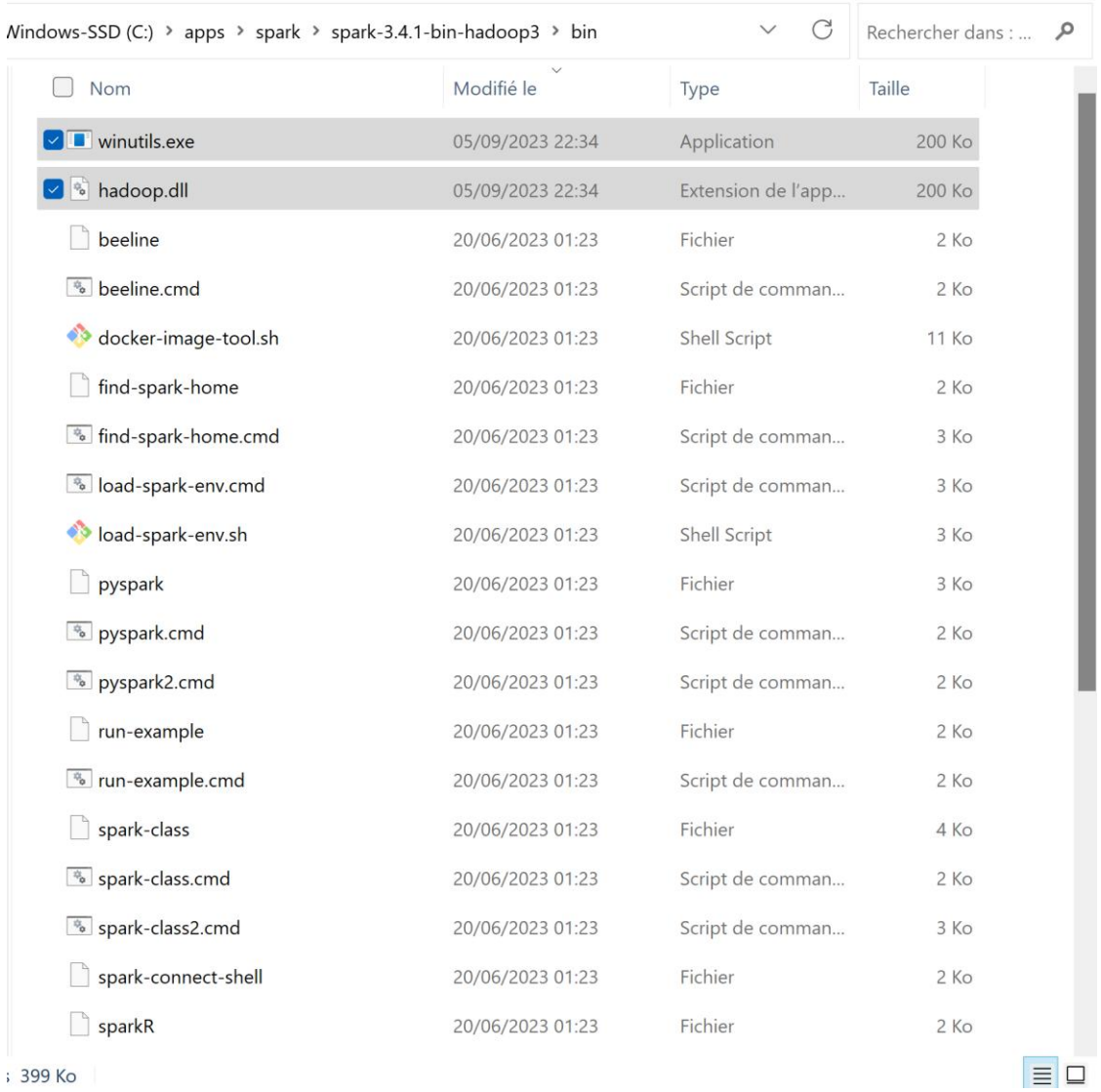
https://github.com/cdarlint/winutils



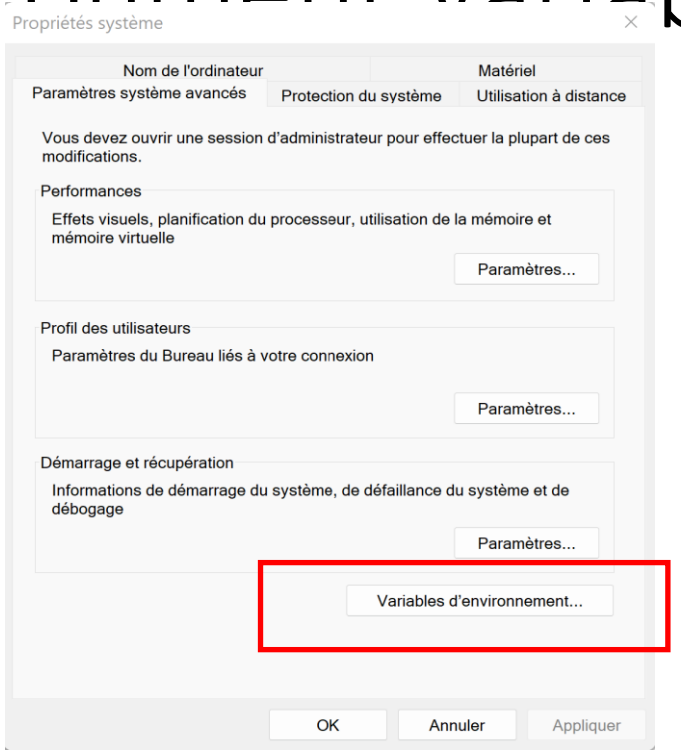
Step 4… Copy winutils.exe, hadoop.dll to you Spark \bin\



Check Copied 2 Files in spark…. \bin



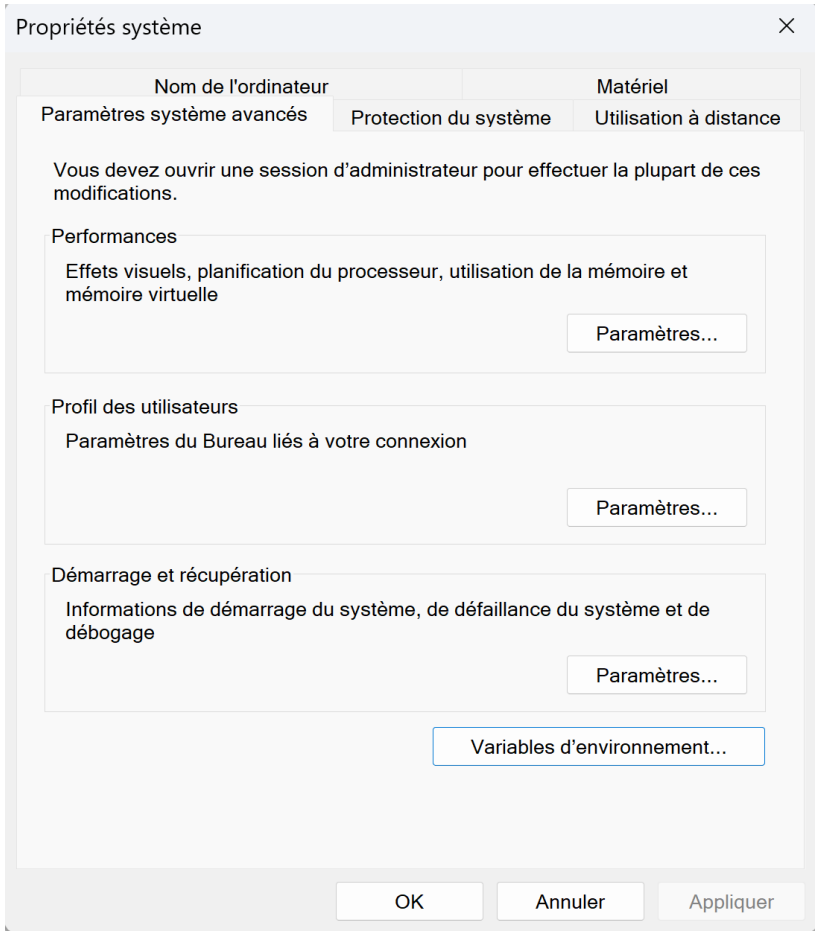
Step 6 : Configure Environment Variables

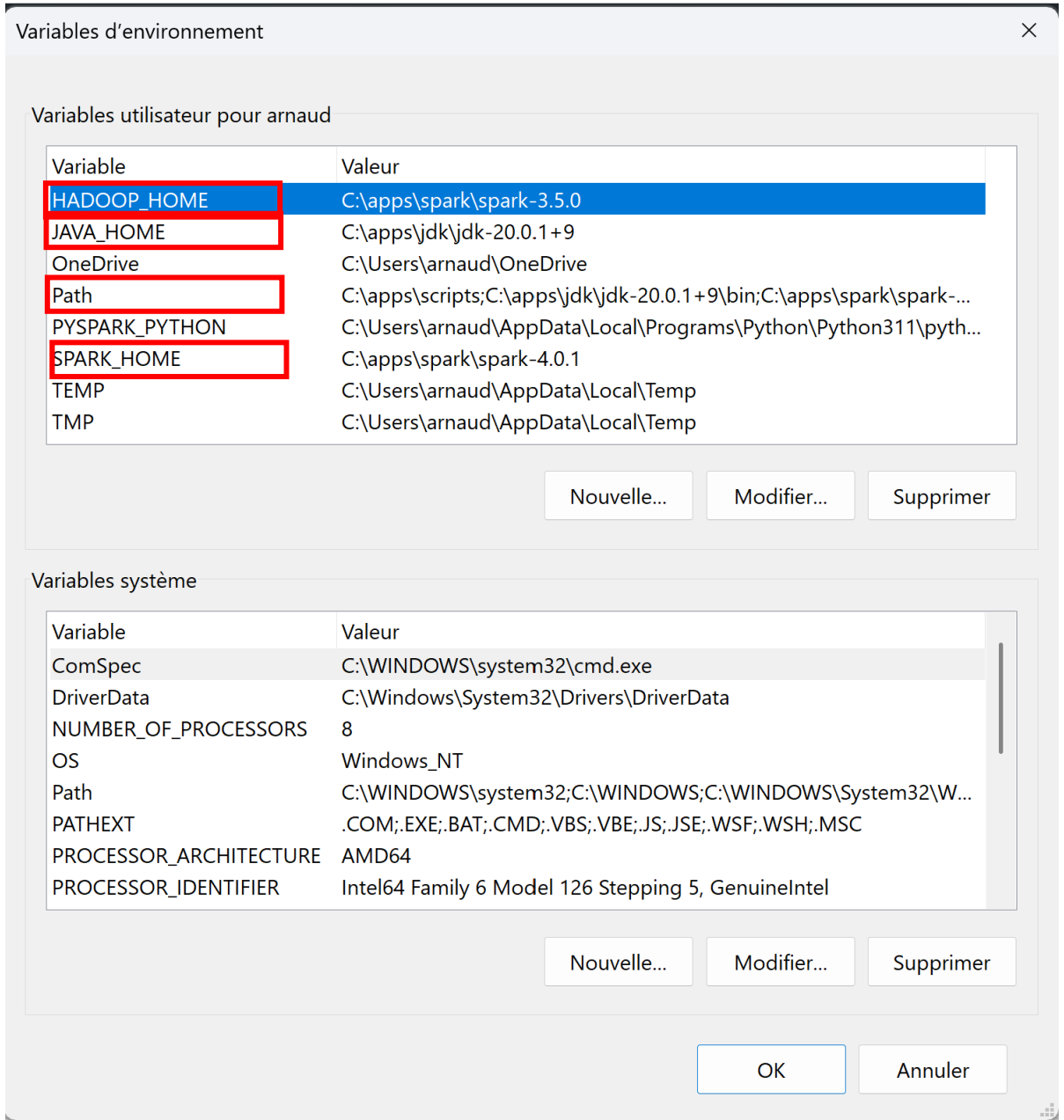


(RECOMMENDED)
Edit your Windows
Environment Variables !!!

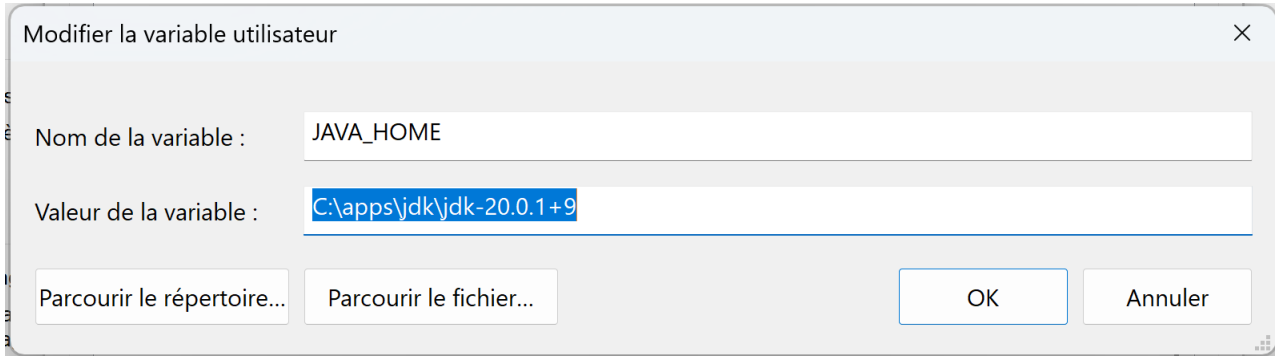
Or save « set VAR=.. » in new file « c:\apps\setenv-spark-xxx.cmd »
This file as to be executed each time you start a terminal !
(typically in « .bashrc » for bash shell or conf in Cmder)

Using Windows Environment Variables





Variable JAVA_HOME

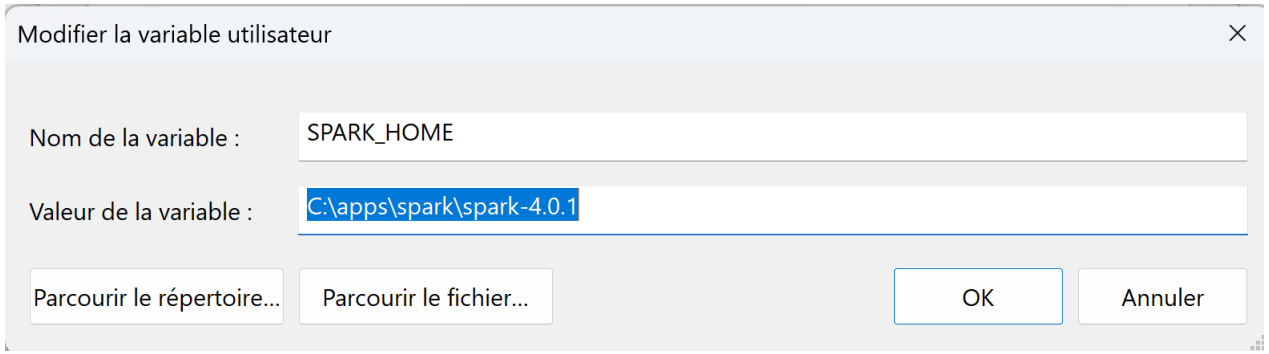


IMPORTANT :
1/ no space in directory path
(so do not use the default « c:\Program & Files\Elipse Adoptium\ … »
=> move to « c:\apps\ » directory

2/ do not use too recent jdk 23 !! It is not compatible with spark 4

3/ recommended JDK: 21 (LTS « Long Term Support », or 19, 17, .. Or « experimental » 22, 20, ..)

SPARK_HOME



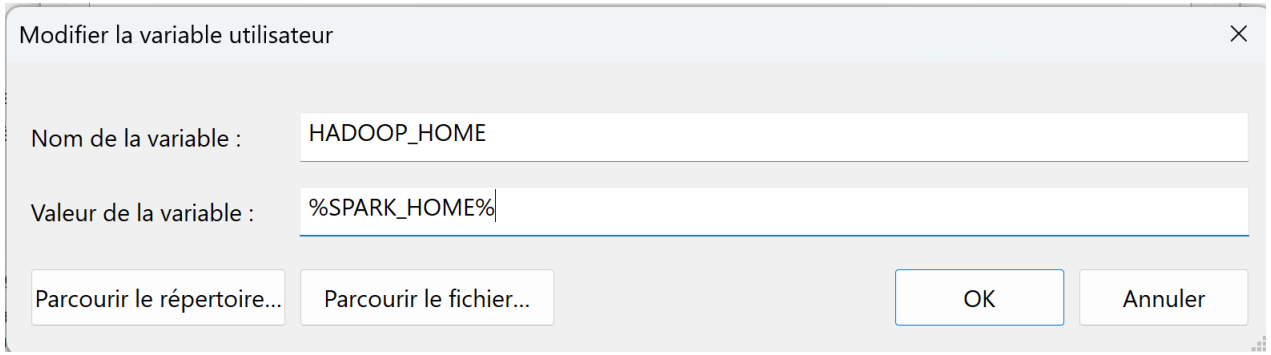
IMPORTANT:
1/ no space in directory path

2/ do not point to parent dir (extracting 2x .tar.gz), nor to child dir (\bin)

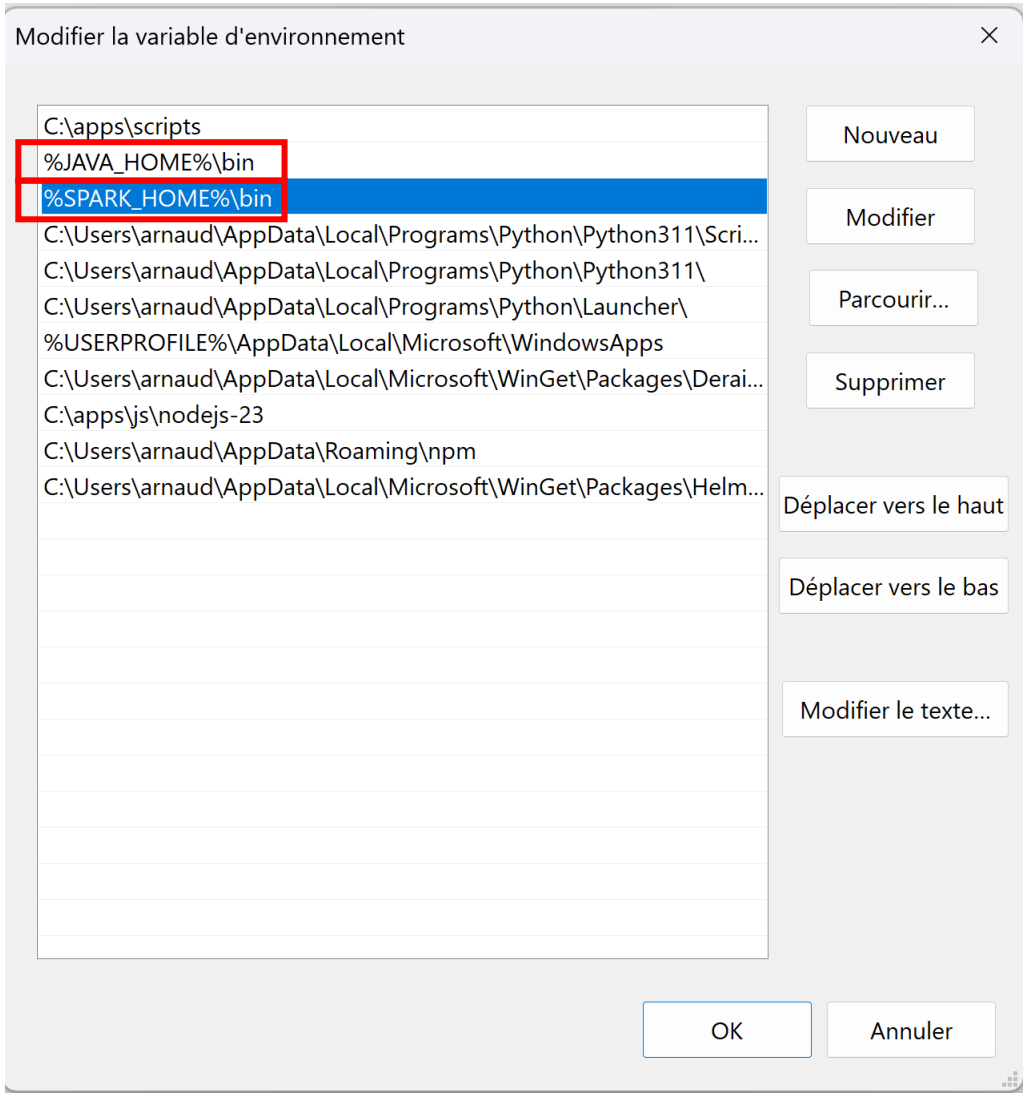
HADOOP_HOME

⇒ points to same dir as SPARK_HOME
(we downloaded spark distribution « with-hadoop-embedded»)

⇒ It is how hadoop library find its config /*.xml,
and also hadoop.dll + winutil.exe files



PATH



Important: DO NOT « corrupt » (delete) your PATH !!! You may have problems on your computer
JUST add 2 « new » entries: %JAVA_HOME%\bin and %SPARK_HOME%\bin
and cleanup old unused java conflicts (like old « oraclepath »)

(NOT Recommended, ALTERNATIVE …) using scripts files to exec set JAVA_HOME , PATH, SPARK_HOME, HADOOP_HOME

Example for OLD version of spark 3.3.0

setenv-spark-td1.cmd

```
@echo off
@echo ... executing setenv-spark-td1.cmd

REM set JAVA_HOME=C:\apps\jdk\jdk-8  ... does not work on windows
set JAVA_HOME=C:\apps\jdk\jdk-17.0.1.12
set PATH=%JAVA_HOME%\bin;%PATH%

REM currently use hadoop jars provided in spark, but could be separated
set HADOOP_HOME=C:\apps\hadoop\spark-3.3.0-hadoop-3.3

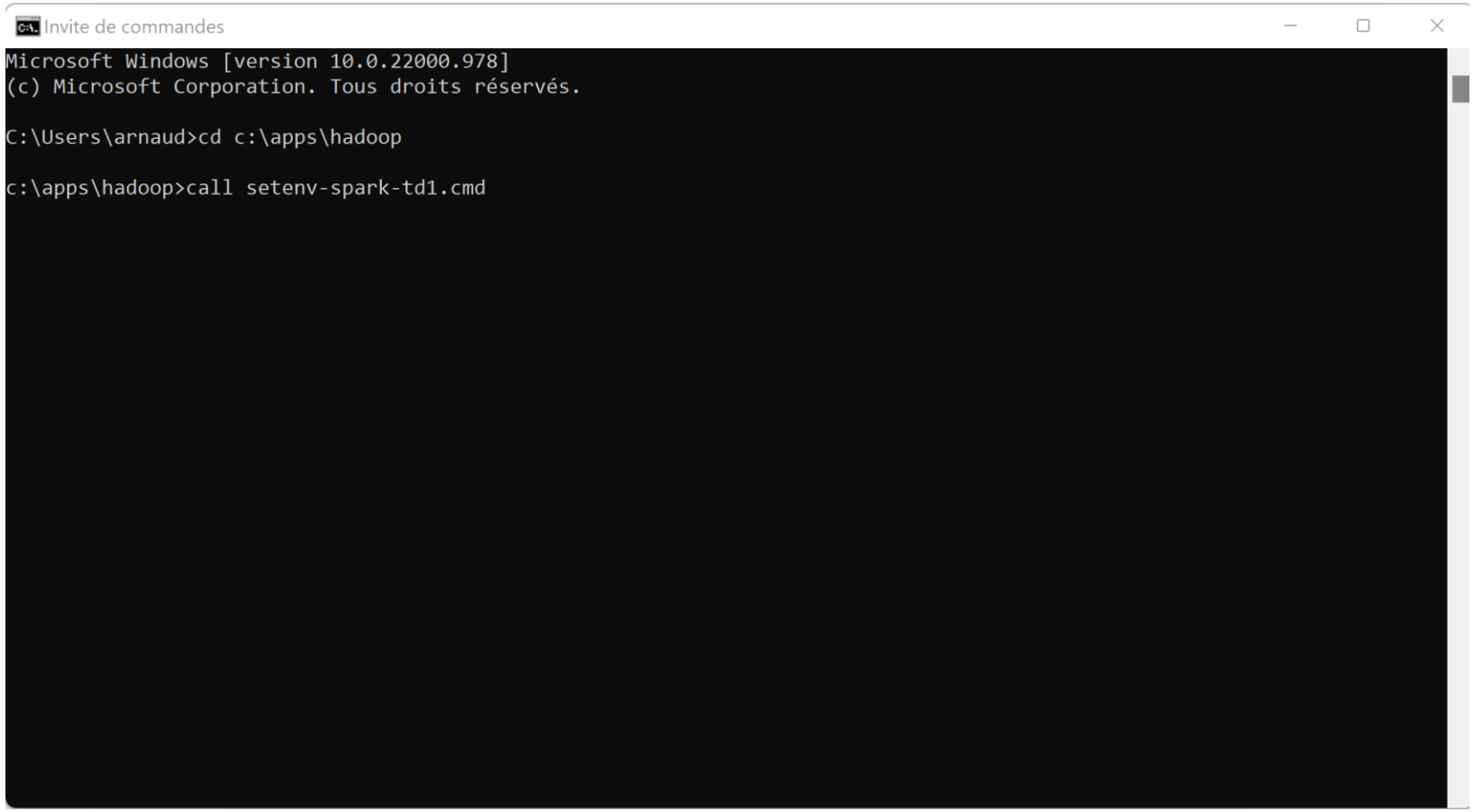
set SPARK_HOME=C:\apps\hadoop\spark-3.3.0-hadoop-3.3
set PATH=%SPARK_HOME%\bin;%PATH%

set SPARK_SCALA_VERSION=2.12

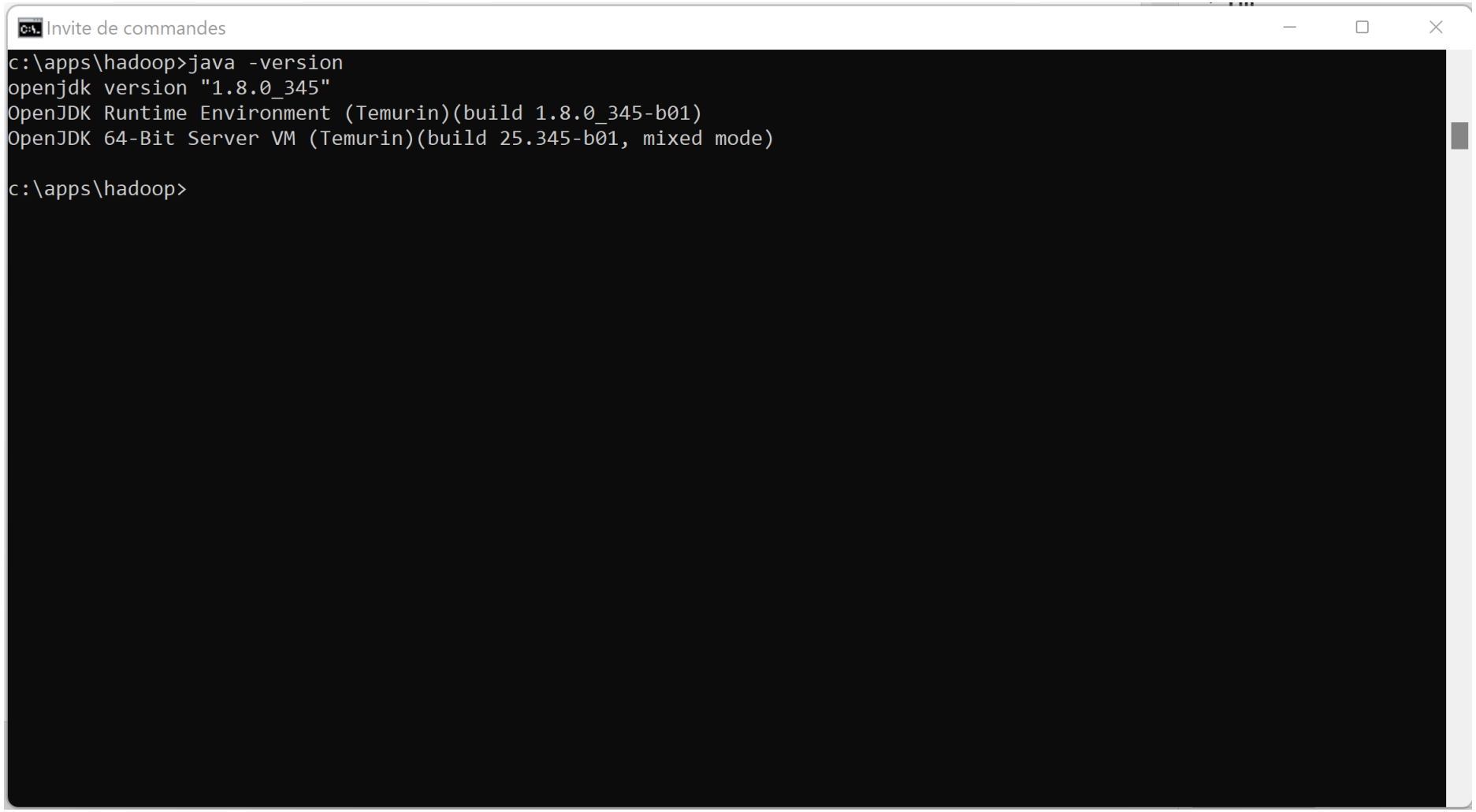
@echo .. using JAVA_HOME: %JAVA_HOME%
@echo .. using HADOOP_HOME: %HADOOP_HOME%
@echo .. using SPARK_HOME: %SPARK_HOME%
@echo .. using SPARK_SCALA_VERSION: %SPARK_SCALA_VERSION%
```

Step 7: Sanity Checks …

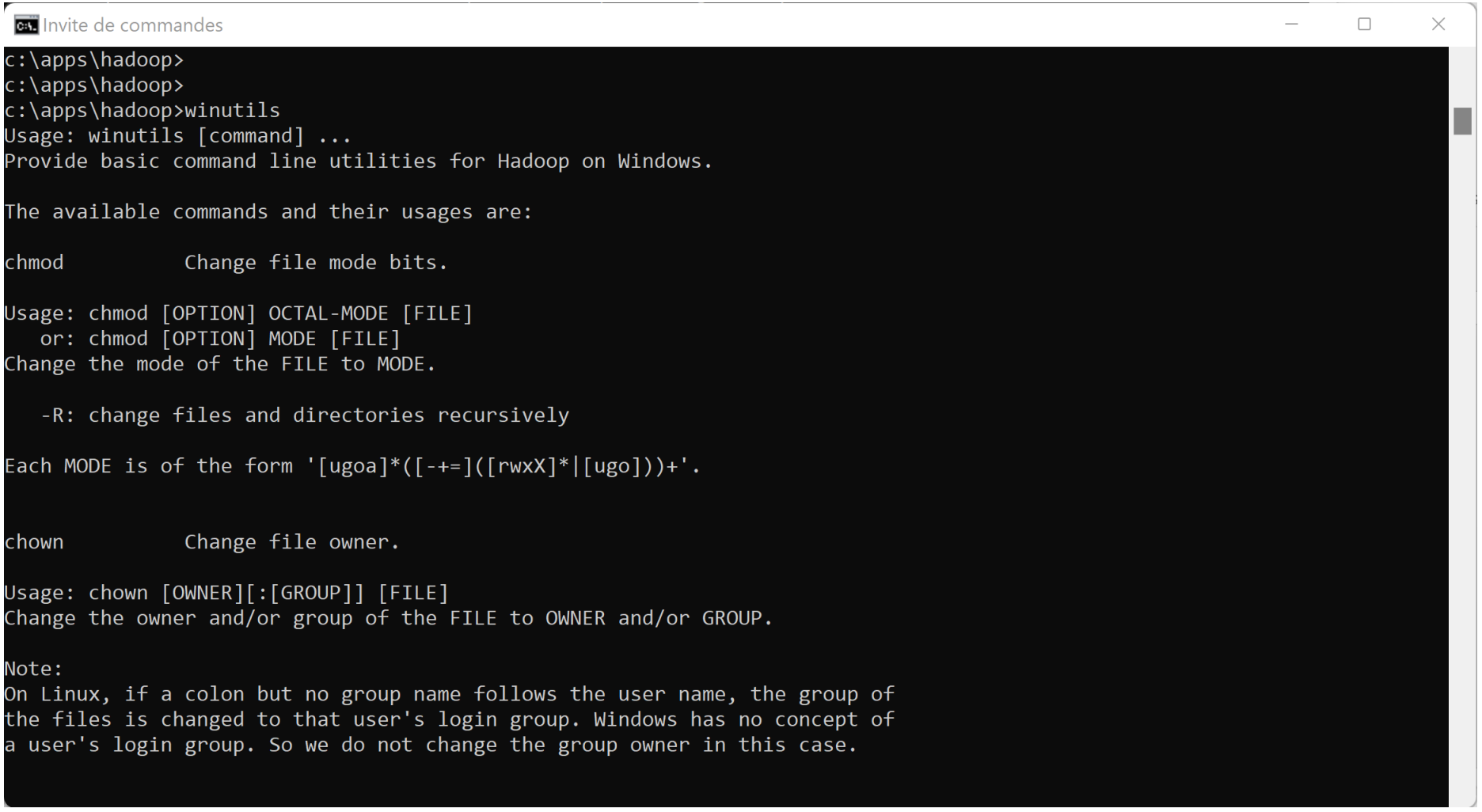
Windows > cmd
cd <yourdir>
call <your-setenv>.cmd



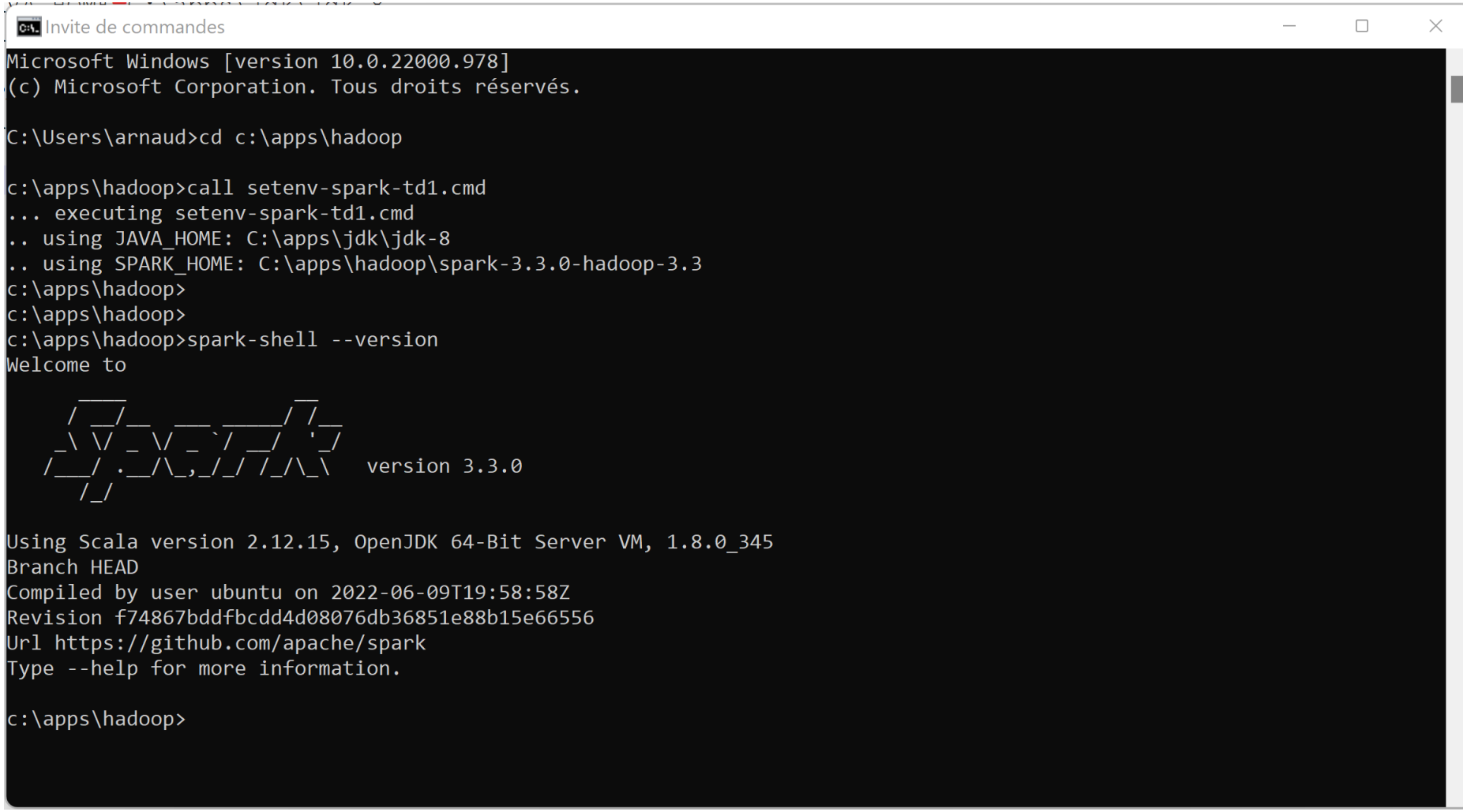
Sanity Check: Java



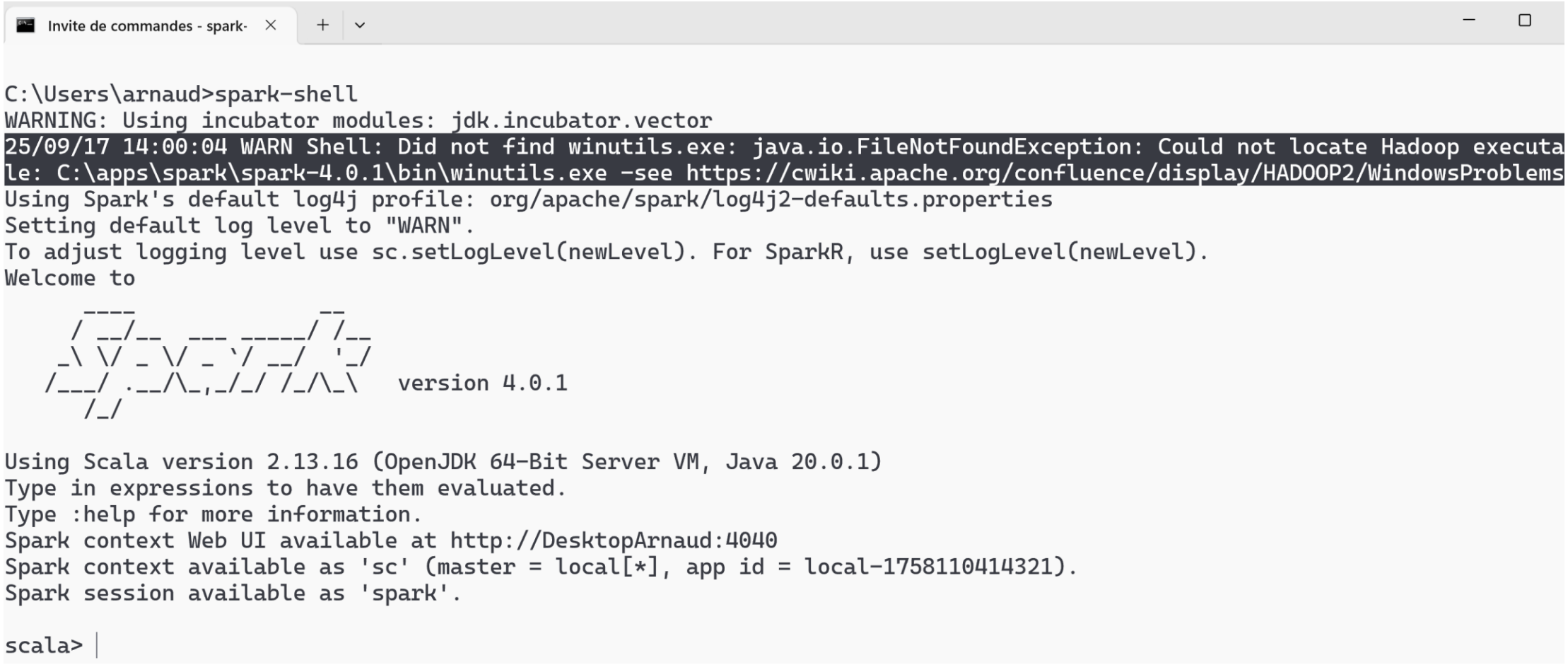
Sanity check: WinUtils



Sanity Check « spark-shell --version »



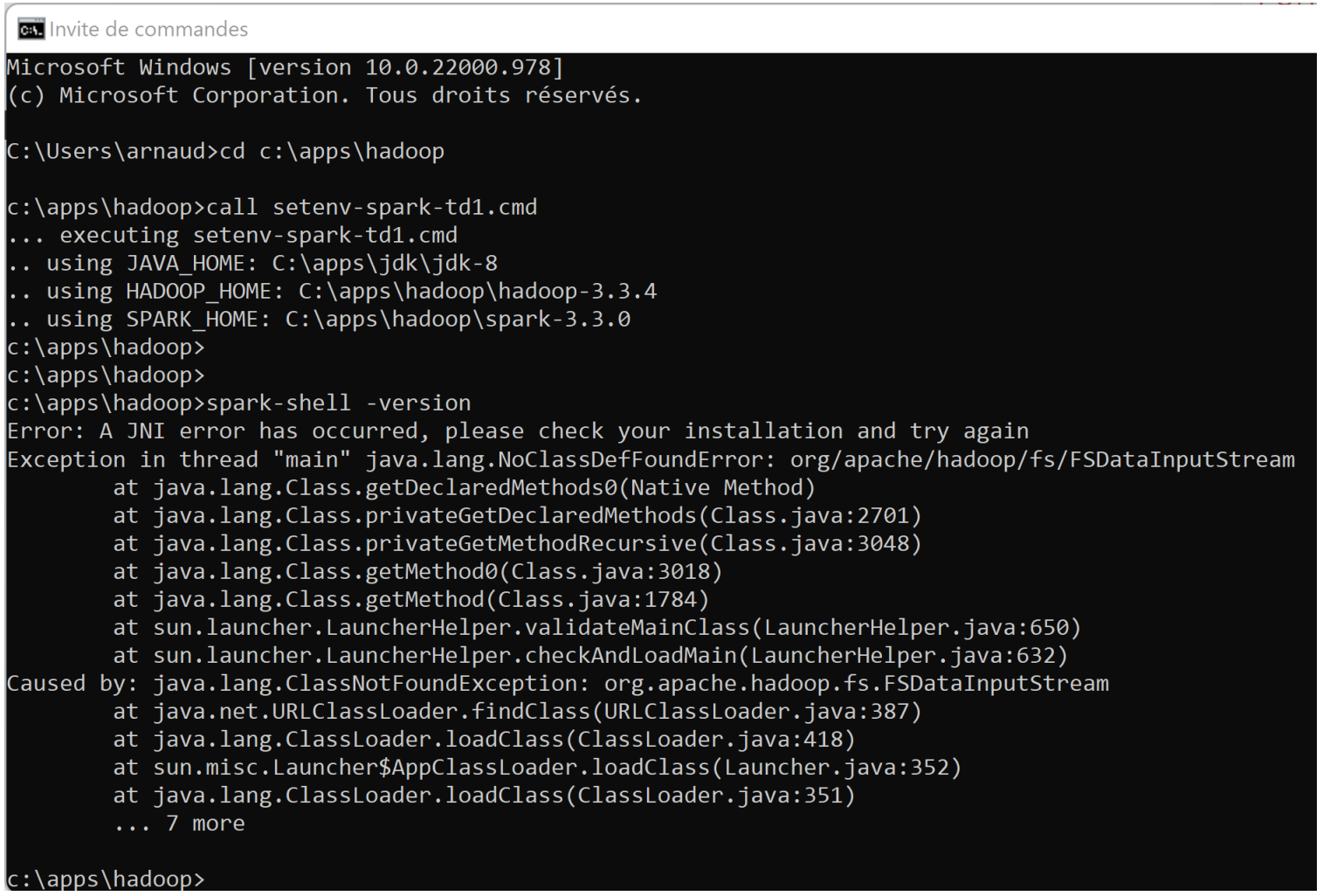
Typical Error: forgot to copy winutils.exe file



Spark and custom user-defined Hadoop?
.. More difficult
error when using JDK8 … but ok when JDK17 ?!

Naively think
missing hadoop jars
in spark classpath ?!
… NOT Only !!

Root cause
was jdk version +
Missing env var
SPARK_SCALA_VERSION



Optionnal when using custom Hadoop Sanity check HADOOP_HOME + PATH

C:> hdfs dfs –ls file:///c:/data

```
C:\apps\hadoop>hdfs dfs -ls file:///c:/data
Found 6 items
d---------   - arnaud Aucun       4096 2022-09-27 16:35 file:///c:/data/OpenData-gouv.fr
----------   1 arnaud Aucun        450 2021-12-31 19:41 file:///c:/data/README.txt
----------   1 arnaud Aucun        450 2021-12-31 19:41 file:///c:/data/loremIpsum.txt
----------   1 arnaud Aucun        371 2022-09-27 14:50 file:///c:/data/sample-data1-no-header.csv
----------   1 arnaud Aucun        405 2022-09-27 13:57 file:///c:/data/sample-data1.csv
-r--r--r--   1 arnaud Aucun    2777803 2022-09-27 15:12 file:///c:/data/sample2.csv

C:\apps\hadoop>dir c:\data
 Le volume dans le lecteur C s'appelle Windows-SSD
 Le numéro de série du volume est 3EFF-0F82

 Répertoire de c:\data

30/09/2022  20:26    <DIR>          .
31/12/2021  20:41               450 loremIpsum.txt
27/09/2022  16:35    <DIR>          OpenData-gouv.fr
31/12/2021  20:41               450 README.txt
27/09/2022  14:50               371 sample-data1-no-header.csv
27/09/2022  13:57               405 sample-data1.csv
27/09/2022  15:12         2 777 803 sample2.csv
               5 fichier(s)        2 779 479 octets
               2 Rép(s)  633 680 625 664 octets libres
```

hdfs dfs –ls …

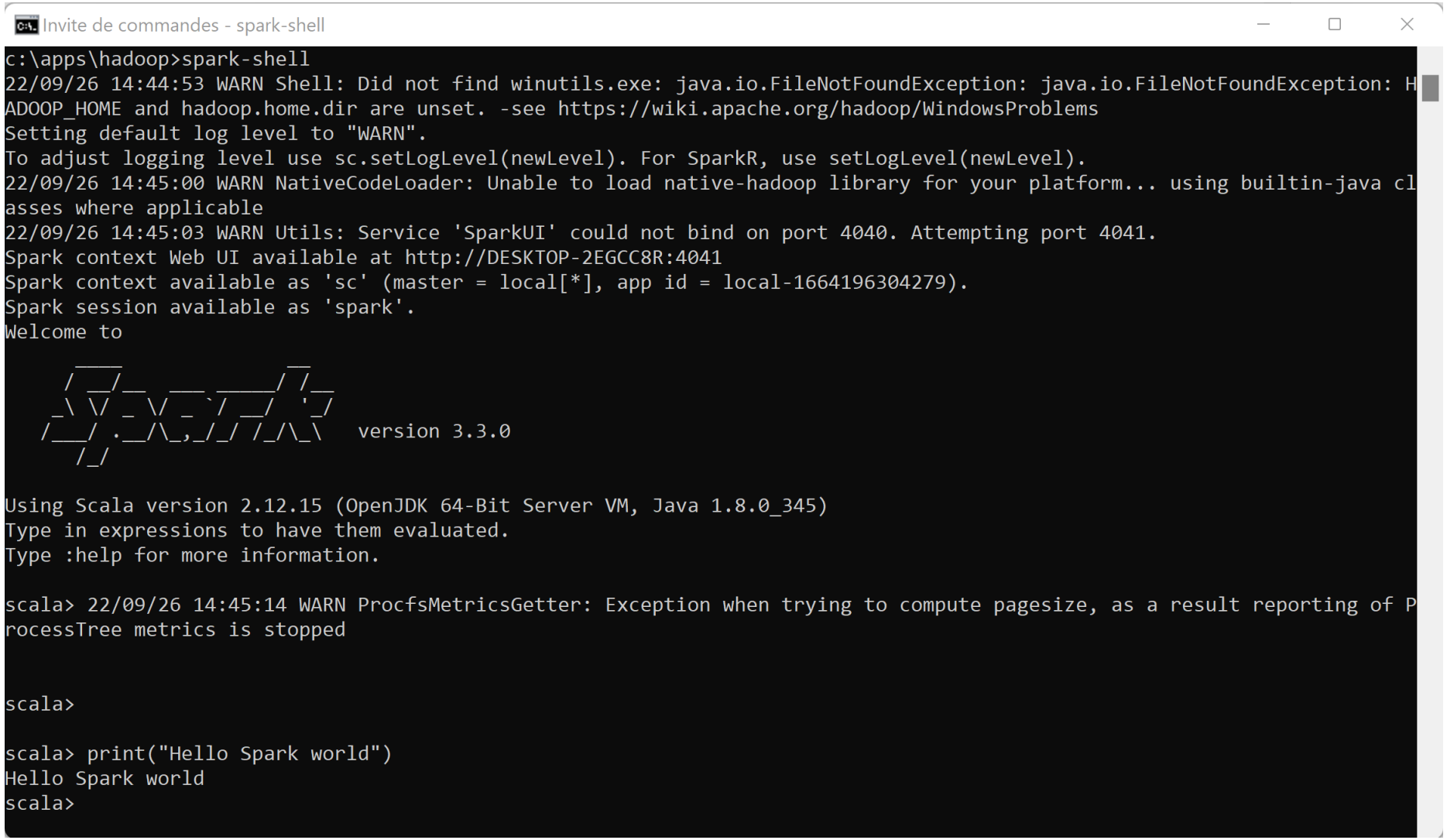
« hdfs dfs » is unix like commands for accessing Hadoop FileSystem

example
hdfs dfs –ls /some-hdfs-dir
    # => implicitly converted to «hdfs dfs –ls hdfs://<hdfsServer>/some-hdfs-dir »

hdfs dfs –ls s3:// … for accessing Amazon S3
hdfs dfs –ls abfss:// … for accessing Azure Storage

hdfs dfs –ls file:///c:/local-dir … for accessing local dir ! Notice 3 slashes

Spark-shell … print(« Hello World »)



Objectives





1/ Install on your (Windows) PC a minimalist local SPARK

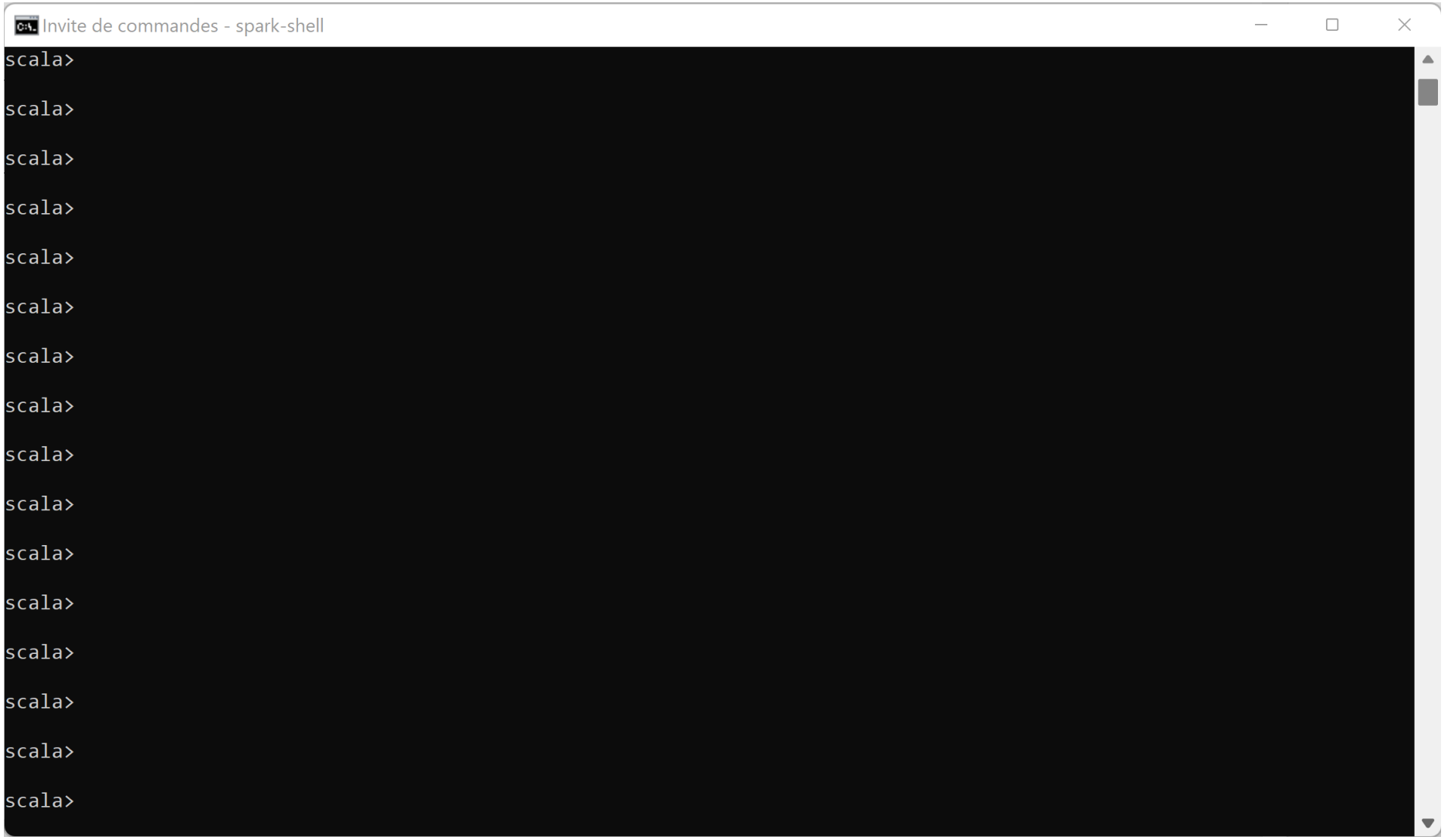
2/ Configure it, launch spark-shell

3/ Discover spark-shell scala> REPL

4/ Execute basic spark commands on DataSets, Files

10 mn Pause

Discover Spark-shell
type <enter> <enter> <enter> …



REPL = Read – Eval – Print - Loop

For each line,

Spark evaluate (compile in scala code)

Result is printed, with type information
and assigned to variable « res${i} »
  where « i » is incremented

Variables can be re-used by name

```
scala> 1
res10: Int = 1

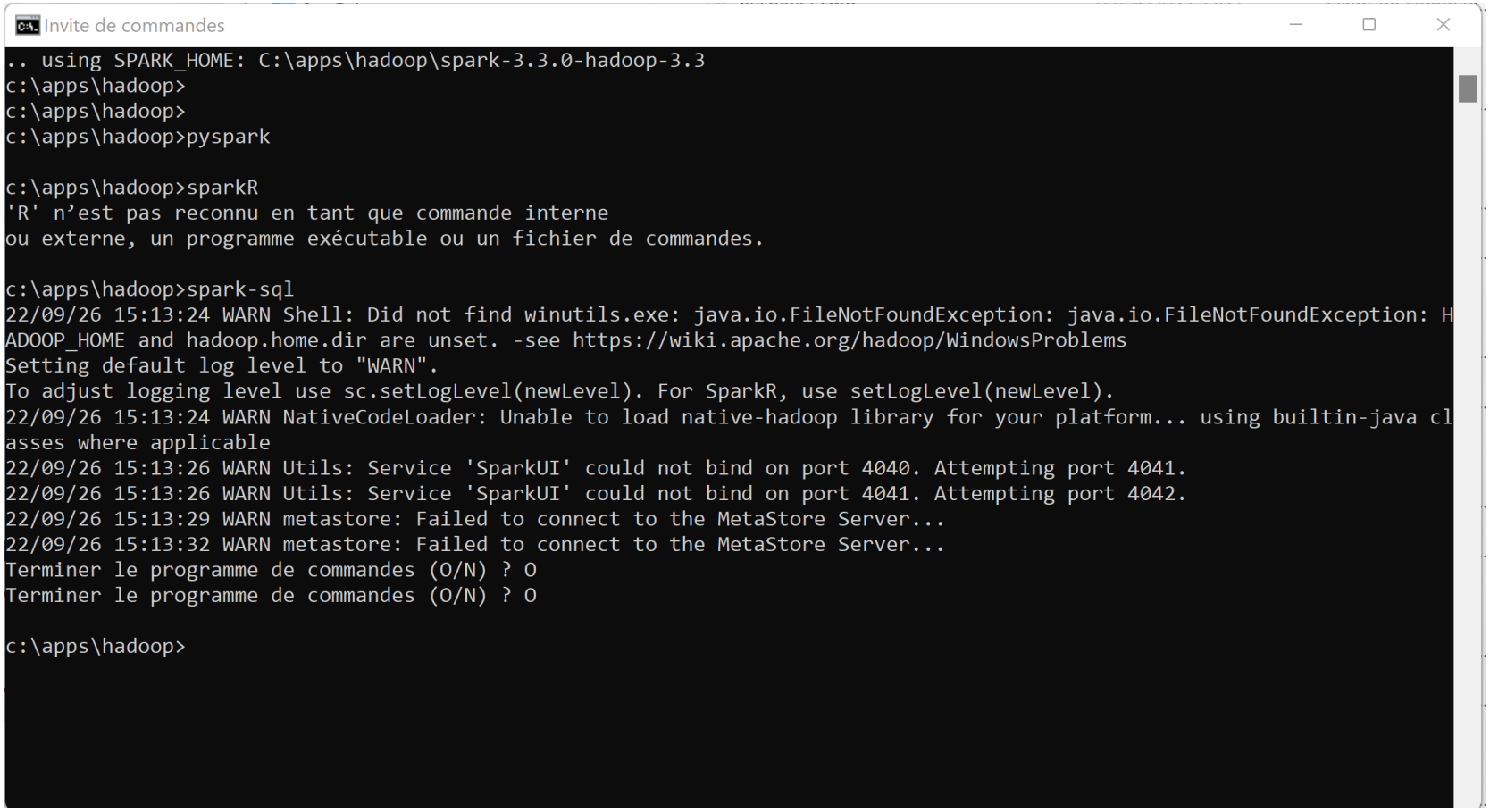
scala> 2
res11: Int = 2

scala> res10 + res11
res12: Int = 3

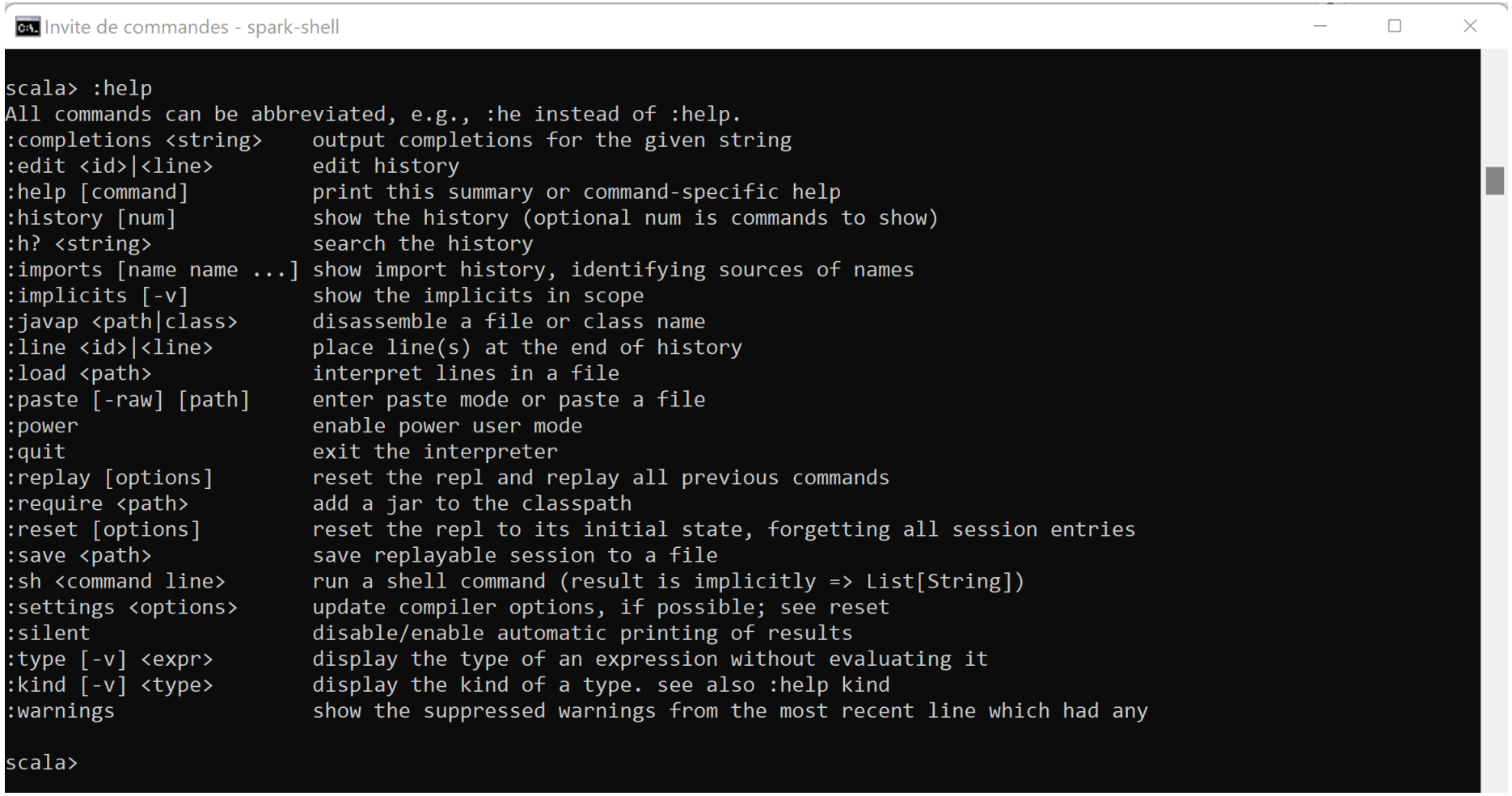
scala> "Whaouh!"
res13: String = Whaouh!

scala>
```

Not only spark-shell…
pyspark => need python exe
sparkR => need ‘R’ exe
spark-sql => need connect to Hive MetaStore Server



Discover built-ins commands, :help



Discover History

Exercise :

a/ Type 5 basic commands:
print(« line1 ») <ENTER>
print(« line2 ») <ENTER>
….
print(« line5 ») <ENTER>

b/ navigate scroll UP and Down to retype command N

c/ replay all commands, using « :replay »

c/ save session in file « my-supper-commands.txt »

Interactive Line Edit
Jline (~GNU readline)

Remember the following standard shortcuts

CTRL+L : clear screen

CTRL+a / CTRL+e : jump to begin/end of line

CTRL+d : delete char
CTRL+k : kill rest of line, put in buffer
CTRL+y : paste line from buffer

Scroll Up / Down : previous/next line from history

Discover History (stop + relaunch spark-shell)

Exercise :

a/ Stop you spark-shell

b/ restart a spark-shell

c/ do you still see your previous commands (Scroll Up)

d/ do you still see your history ?

e/ Find in which file are written your commands ?
… search for hidden « .scala_history » file, in your home directory C:\users\<login>\.scala_history

Exercise using « :save », « :load »

a/ save all commands session in file « my-super-commands.txt »

b/ load and re-execute commands from file

Exercise using « :replay », « :reset »

a/ replay all in once your previous commands, using « :replay »

b/ reset commands… see History after

Spark-shell is a « REPL » for Spark in « scala »

Scala basic commands …

```
scala> var x = 5
x: Int = 5

scala> print("x:" + x)
x:5
scala> print(s"x: ${x}")
x: 5
```

More Scala …

```
scala> (1 to 3).foreach(x => println(x))
1
2
3
```

More Scala with … Sequence, implicit Lambda, « _ » var

```
scala> var ls=Seq(1, 3, "Hello")
ls: Seq[Any] = List(1, 3, Hello)

scala> ls.foreach(println(_))
1
3
Hello
```

« Scala » .. ~ a super set of « Java »
Scala runs on the JVM

```
scala> java.lang.System.out.println("Hello plain old java from Spark-scala")
Hello plain old java from Spark-scala
```

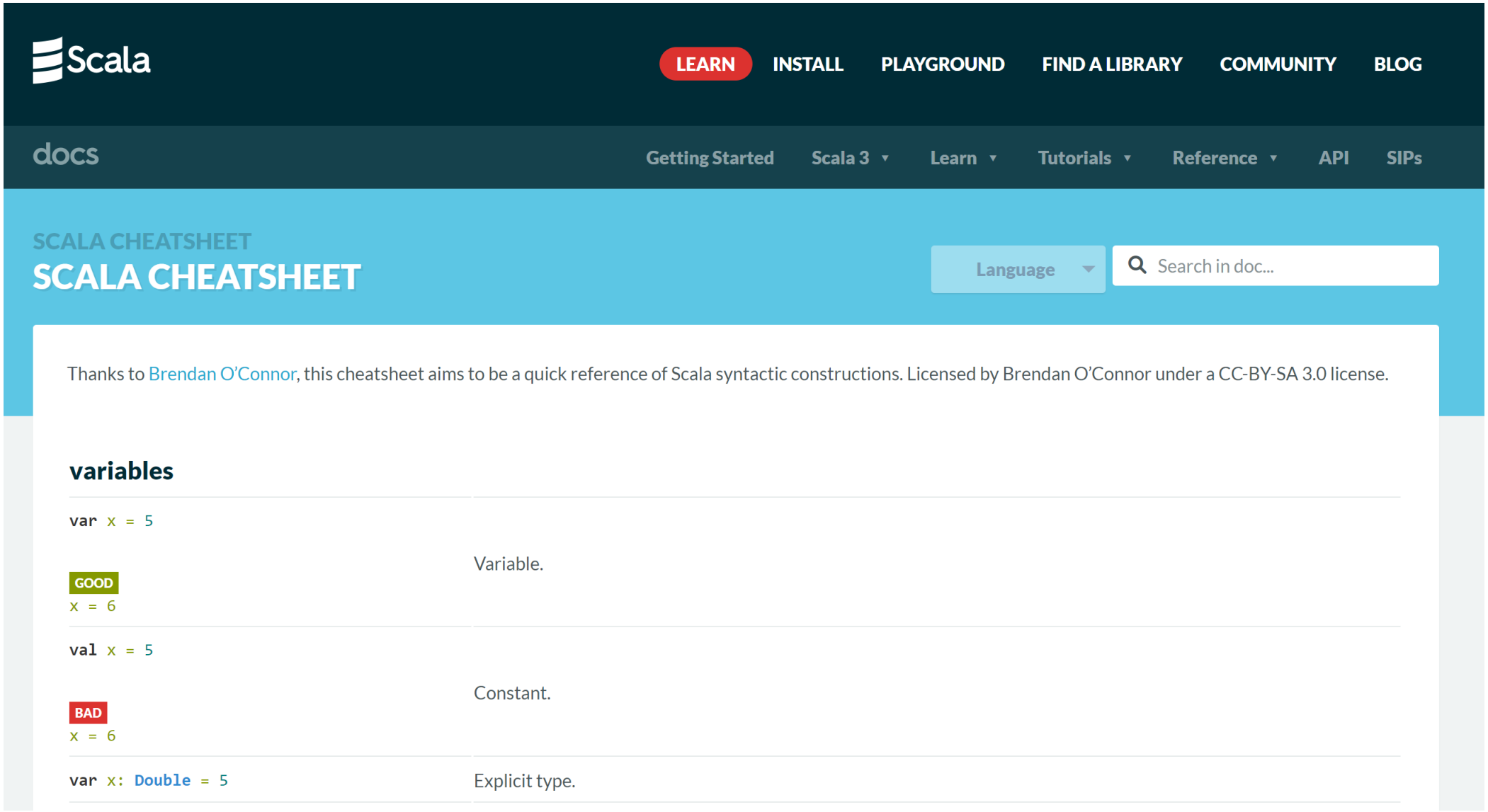
… but not exactly

```
scala> for(int i=0; i < 3; i++) System.out.println("hello " + i)
<console>:1: error: illegal start of simple pattern
       for(int i=0; i < 3; i++) System.out.println("hello " + i)
                  ^
<console>:1: error: '<-' expected but ';' found.
```

```
scala> for(i <- 0 to 2) System.out.println("hello " + i)
hello 0
hello 1
hello 2
```

https://docs.scala-lang.org/cheatsheets/

5-10 minutes
Reading



Objectives



1/ Install on your (Windows) PC a minimalist local SPARK



2/ Configure it, launch spark-shell



3/ Discover spark-shell scala> REPL

4/ Execute basic spark commands on DataSets

Introduction to Exercises 1,2,3,4,5 …
The goal is to create a spark DataSet..

Expected DataSet = List of 100_000 rows
each row containing a Tuple of
(int : i,
boolean : true if i is even,
string: « hello ${i} »)

Exercise 1 : create a List of 3 Tuples

var ls = ???

With expected content:

(1, false, "hello 1")
(2, true, "hello 2")
(3, false, "Hello 3")

Hint Exercise 1

In scala,

« (a, b, c) » creates a Tuple with 3 fields _1,_2,_3 with _1=a, _2=b, _3=c
« Seq(a, b, c) » create a Sequence (synonym: List) of 3 elements: a, b, c

Answer Exercise 1

```
scala> var ls = Seq( (1, false, "hello1"), (2, true, "hello 2"), (3, false, "Hello 3"))
ls: Seq[(Int, Boolean, String)] = List((1,false,hello1), (2,true,hello 2), (3,false,Hello 3))
```

Exercise 2: same, but create an empty List, and successively add 3 rows … the (non idiomatic) Java way

Hint exercise 2

List are immutable !
You can create a new List, as the concatenation of 1 element and an existing list
List has « :: » operator, not Seq !

```
var ls = List[Object]()
var ls2 =  a :: ls
```

« a » and « ls » must have same generic type!
Use « List[Type] » instead of « List() »
Type is … (Int, Boolean, String)

Answer Exercise 2 … the (non-idiomatic) Java Way..

```
scala> var ls = List[(Int,Boolean,String)]()
ls: List[(Int, Boolean, String)] = List()

scala> ls = (1, false, "hello " +1) :: ls
ls: List[(Int, Boolean, String)] = List((1,false,hello 1))

scala> ls = (2, true, "hello " +2) :: ls
ls: List[(Int, Boolean, String)] = List((2,true,hello 2), (1,false,hello 1))

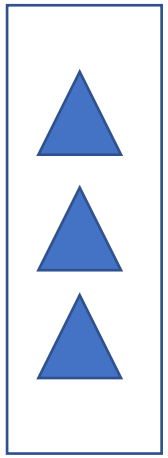
scala> ls = (3, false, "hello " +3) :: ls
ls: List[(Int, Boolean, String)] = List((3,false,hello 3), (2,true,hello 2), (1,false,hello 1))
```

Exercise 3 .. Same using « functional » code style: transform the sequence (1, 2 … 10000) into Tuples using « map » lambda function

Hint Exercise 3

« .map(f) » is a method of List,
to map all elements from source type X to target type Y
taking a function « f » as parameter : « f(X) -> Y »
… f can be written as a lambda: « x => { ..return y;} »

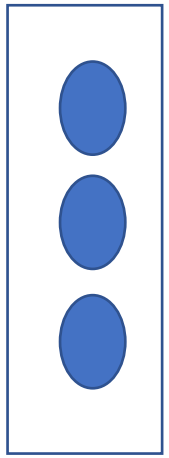
Source : List[X]



transformationFunction : X -> Y



target : List[Y]



Answer Exercise 3

```
scala> var ls = (1 to 100000).map(i => (i, i%2 == 0, s"Hello ${i}") )
ls: scala.collection.immutable.IndexedSeq[(Int, Boolean, String)] = Vector((1,false,Hello 1), (2,true,Hello 2), (3,false
,Hello 3), (4,true,Hello 4), (5,false,Hello 5), (6,true,Hello 6), (7,false,Hello 7), (8,true,Hello 8), (9,false,Hello 9)
, (10,true,Hello 10), (11,false,Hello 11), (12,true,Hello 12), (13,false,Hello 13), (14,true,Hello 14), (15,false,Hello
15), (16,true,Hello 16), (17,false,Hello 17), (18,true,Hello 18), (19,false,Hello 19), (20,true,Hello 20), (21,false,Hel
lo 21), (22,true,Hello 22), (23,false,Hello 23), (24,true,Hello 24), (25,false,Hello 25), (26,true,Hello 26), (27,false,
Hello 27), (28,true,Hello 28), (29,false,Hello 29), (30,true,Hello 30), (31,false,Hello 31), (32,true,Hello 32), (33,fal
se,Hello 33), (34,true,Hello 34), (35,false,Hello 35), (36,true,Hello 36), (3...
```

Exercise 4: display result from previous exercise

4a / display length of result
4b/ display row 0 tuple
4c/ display row 0, individual fields from tuple

Answer Exercise 4

```
scala> ls.size
res44: Int = 100000

scala> ls(0)
res45: (Int, Boolean, String) = (1,false,Hello 1)

scala> ls(0)._1
res46: Int = 1

scala> ls(0)._2
res47: Boolean = false

scala> ls(0)._3
res48: String = Hello 1
```

Exercise 5: create a spark DataSet of 100_000 rows
each row containing a Tuple of
int : i
boolean : true if i is even
string: « hello ${i} »

Hint Exercise 5

Just use

spark.createDataset() method

« spark » is a built-in variable
You can type « spark.create » and press TAB TAB to autocomplete

Answer Exercise 5

```
scala> var ls = (1 to 100000).map(i => (i, i%2 == 0, s"Hello ${i}") )
ls: scala.collection.immutable.IndexedSeq[(Int, Boolean, String)] = Vector((1,false,Hello 1), (2,true,Hello 2), (3,false
,Hello 3), (4,true,Hello 4), (5,false,Hello 5), (6,true,Hello 6), (7,false,Hello 7), (8,true,Hello 8), (9,false,Hello 9)
, (10,true,Hello 10), (11,false,Hello 11), (12,true,Hello 12), (13,false,Hello 13), (14,true,Hello 14), (15,false,Hello
15), (16,true,Hello 16), (17,false,Hello 17), (18,true,Hello 18), (19,false,Hello 19), (20,true,Hello 20), (21,false,Hel
lo 21), (22,true,Hello 22), (23,false,Hello 23), (24,true,Hello 24), (25,false,Hello 25), (26,true,Hello 26), (27,false,
Hello 27), (28,true,Hello 28), (29,false,Hello 29), (30,true,Hello 30), (31,false,Hello 31), (32,true,Hello 32), (33,fal
se,Hello 33), (34,true,Hello 34), (35,false,Hello 35), (36,true,Hello 36), (3...

scala> val ds = spark.createDataset(ls)
ds: org.apache.spark.sql.Dataset[(Int, Boolean, String)] = [_1: int, _2: boolean ... 1 more field]
```

Equivalent 1-liner solution :

```
scala> val dsDirect = spark.createDataset( (1 to 100000).map(i => (i, i%2 == 0, s"Hello ${i}") ) )
dsDirect: org.apache.spark.sql.Dataset[(Int, Boolean, String)] = [_1: int, _2: boolean ... 1 more field]
```

Exercise 6: Display result DataSet of ex. 5

6a / display number of rows
6b/ show first rows (default: 20)
6c/ show only first 2 rows
6d/ show individual columns of row0

Hint Exercise 6

Like on Seq or List … Dataset accept many methods,
like « .show() », « .count() », « .take() » etc.

Answer Exercise 6

```
scala> ds.count
res49: Long = 100000

scala> ds.show(2)
+---+-----+-------+
| _1|   _2|     _3|
+---+-----+-------+
|  1|false|Hello 1|
|  2| true|Hello 2|
+---+-----+-------+
only showing top 2 rows
```

```
scala> ds.take(1)(0)
res57: (Int, Boolean, String) = (1,false,Hello 1)

scala> ds.take(1)(0)._1
res58: Int = 1

scala> ds.take(1)(0)._2
res59: Boolean = false

scala> ds.take(1)(0)._3
res60: String = Hello 1
```

By default… « .show() » => same as « .show(20) »
.. => same as « .show(20, true) »

```
scala> ds.show()
+---+-----+--------+
| _1|   _2|      _3|
+---+-----+--------+
|  1|false| Hello 1|
|  2| true| Hello 2|
|  3|false| Hello 3|
|  4| true| Hello 4|
|  5|false| Hello 5|
|  6| true| Hello 6|
|  7|false| Hello 7|
|  8| true| Hello 8|
|  9|false| Hello 9|
| 10| true|Hello 10|
| 11|false|Hello 11|
| 12| true|Hello 12|
| 13|false|Hello 13|
| 14| true|Hello 14|
| 15|false|Hello 15|
| 16| true|Hello 16|
| 17|false|Hello 17|
| 18| true|Hello 18|
| 19|false|Hello 19|
| 20| true|Hello 20|
+---+-----+--------+
only showing top 20 rows
```

Exercise 7: Rename columns

Rename columns _1,_2,_3
to
« id », « even », « text »

Hint Exercise 7

Use dataset.withColumnRenamed()

… remember Dataset are immutable !
Most Dataset method returns a new Dataset

so cascade calls (or assign intermediate results)
repeat the operation 3 times

Answer Exercise 7

```
scala> val ds2 = ds.withColumnRenamed("_1", "id")
ds2: org.apache.spark.sql.DataFrame = [id: int, _2: boolean ... 1 more field]

scala> val ds3 = ds2.withColumnRenamed("_2", "even")
ds3: org.apache.spark.sql.DataFrame = [id: int, even: boolean ... 1 more field]

scala> val ds4 = ds3.withColumnRenamed("_3", "text")
ds4: org.apache.spark.sql.DataFrame = [id: int, even: boolean ... 1 more field]
```

Ds … is unmodified … still has _1, _2, _3 columns

```
scala> ds
res62: org.apache.spark.sql.Dataset[(Int, Boolean, String)] = [_1: int, _2: boolean ... 1 more field]
```

More idiomatic: cascade methods

```
scala> val dsRenamed = ds.withColumnRenamed("_1", "id").withColumnRenamed("_2", "even").withColumnRenamed("_3", "text")
dsRenamed: org.apache.spark.sql.DataFrame = [id: int, even: boolean ... 1 more field]
```

Exercise 8

Filter rows where « id » is « >= 100 »
Using « filter(Column) » operators

To create an expression on column « id » of dataset « ds »,
You can use « ds(« id ») »
OR implicit equivalent $«id »

Hint Exercise 8

Type « .filter» and pres TAB TAB to autocomplete
You can see there are 3 possible overloads for method

```
scala> val dsFilter = ds4.filter

def filter(func: org.apache.spark.api.java.function.FilterFunction[org.apache.spark.sql.Row]): org.apache.spark.sql.Data
set[org.apache.spark.sql.Row]
def filter(func: org.apache.spark.sql.Row => Boolean): org.apache.spark.sql.Dataset[org.apache.spark.sql.Row]
def filter(conditionExpr: String): org.apache.spark.sql.Dataset[org.apache.spark.sql.Row]
def filter(condition: org.apache.spark.sql.Column): org.apache.spark.sql.Dataset[org.apache.spark.sql.Row]
```

use « .filter(ds(« columnName») operator…) »

It is more complex than using SQL conditionExpr… cf exercise 9
But It is simpler than exercise 10 … using lambda predicateFunc : row -> boolean

Answer Exercise 8

```
scala> val dsFilter = ds4.filter(ds4("id") >= 100)
dsFilter: org.apache.spark.sql.Dataset[org.apache.spark.sql.Row] = [id: int, even: boolean ... 1 more field]

scala> dsFilter.count
res63: Long = 99901
```

utlisation de l'api BAS-NIVEAU "Column" expression
val ds5 = dataframe.filter(dataframe.col("value").gt(lit (100)))
val ds5 = dataframe.filter(dataframe.col("value") > lit (100))
val ds5 = dataframe.filter(dataframe.col("value") > 100)
val ds5 = dataframe.filter(col("value") > 100)
val ds5 = dataframe.filter($"value" > 100)

Exercise 8

Filter rows where

« id » is « >= 100 »

AND

«text » contains substring « 12 »

Hint Exercise 8

Spark as overriden most of the natural operators on class « Column »
Including >=, &&, ||, !, contains, …

You can manipulate columns objects almost as values in scala

Answer Exercise 8

```
scala> val dsFilter = ds4.filter( ds4("id") >= 100 && ds4("text").contains("12") )
dsFilter: org.apache.spark.sql.Dataset[org.apache.spark.sql.Row] = [id: int, even: boolean ... 1 more field]

scala> dsFilter.show(10)
+---+-----+---------+
| id| even|     text|
+---+-----+---------+
|112| true|Hello 112|
|120| true|Hello 120|
|121|false|Hello 121|
|122| true|Hello 122|
|123|false|Hello 123|
|124| true|Hello 124|
|125|false|Hello 125|
|126| true|Hello 126|
|127|false|Hello 127|
|128| true|Hello 128|
+---+-----+---------+
only showing top 10 rows
```

Alternative Answer Exercise 8

```
scala> val dsFilter = ds4.filter( $"id" >= 100 && $"text".contains("12") )
dsFilter: org.apache.spark.sql.Dataset[org.apache.spark.sql.Row] = [id: int, even: boolean ... 1 more field]

scala> dsFilter.show(10)
+---+-----+---------+
| id| even|     text|
+---+-----+---------+
|112| true|Hello 112|
|120| true|Hello 120|
|121|false|Hello 121|
|122| true|Hello 122|
|123|false|Hello 123|
|124| true|Hello 124|
|125|false|Hello 125|
|126| true|Hello 126|
|127|false|Hello 127|
|128| true|Hello 128|
+---+-----+---------+
only showing top 10 rows
```

Exercise 9

Redo exercise 7,8 using filter by SQL text expression

Answer Exercise 9

```
scala> val dsFilter = ds4.filter( "id >= 100  AND text LIKE '%12%' " )
dsFilter: org.apache.spark.sql.Dataset[org.apache.spark.sql.Row] = [id: int, even: boolean ... 1 more field]

scala> dsFilter.count
res71: Long = 3969

scala> dsFilter.show(10)
+---+-----+---------+
| id| even|     text|
+---+-----+---------+
|112| true|Hello 112|
|120| true|Hello 120|
|121|false|Hello 121|
|122| true|Hello 122|
|123|false|Hello 123|
|124| true|Hello 124|
|125|false|Hello 125|
|126| true|Hello 126|
|127|false|Hello 127|
|128| true|Hello 128|
+---+-----+---------+
only showing top 10 rows
```

Exercise 10

Same as exercise 7,8,9… but using « .filter(row => …) »
  i.e. Filter with lambda function on row

Use getAs[Type](« columnName ») to extract the value of a column, and cast

Hint Exercise 10

Method signature is
.filter((or.apache.spark.sql.Row row) => boolean)

You need to extract column value for object « Row », and interpret as Int or String

for column « id » as Int … use « row.getAs[Int]("id") »
For column « text » as String … use « row.getAs[String](« text") »

Answer Exercise 10

```
scala> val dsFilter = ds4.filter(row => ( row.getAs[Int]("id") >= 100 && row.getAs[String]("text").contains("12") ) )
dsFilter: org.apache.spark.sql.Dataset[org.apache.spark.sql.Row] = [id: int, even: boolean ... 1 more field]

scala> dsFilter.count
res74: Long = 3969

scala> dsFilter.show(10)
+---+-----+---------+
| id| even|     text|
+---+-----+---------+
|112| true|Hello 112|
|120| true|Hello 120|
|121|false|Hello 121|
|122| true|Hello 122|
|123|false|Hello 123|
|124| true|Hello 124|
|125|false|Hello 125|
|126| true|Hello 126|
|127|false|Hello 127|
|128| true|Hello 128|
+---+-----+---------+
only showing top 10 rows
```

Exercise 11 : same as exercise 1 (on Tuple), but using your own custom « case-class »

Declare a scala case-class (also named Bean, POJO or Record in java)
To replace the Tuple « (Int,boolean,String) » of exercise 1

Should be equivalent to java class:

```
public class CustomBean {
    public int id;
    public boolean even;
    public String text;

    public CustomBean(int id, boolean even, String text) {
      this.id = id;  this.even = even;   this.text = text;
   }
}
```

Answer Exercise 11

```
scala> case class CustomBean(id: Int, even: Boolean, text: String) {}
defined class CustomBean

scala> val b = CustomBean(1, false, "Hello 1")
b: CustomBean = CustomBean(1,false,Hello 1)

scala> val ls=Seq(b, YourCustomBean(2, true, "Hello 2"), YourCustomBean(3, false, "Hello 3"))
ls: Seq[Product with Serializable] = List(CustomBean(1,false,Hello 1), YourCustomBean(2,true,Hello 2), YourCustomBean(3,
false,Hello 3))
```

Exercise 12 : Same as Exercise 5,6,7,8,9 .. Using CaseClass

https://spark.apache.org/docs/latest/sql-getting-started.html#creating-datasets

a/ create a List of 100 000 CustomBean objects, using map + lambda function

b/ create a Dataset[CustomBean] with 100 000 rows

c/ display results : show 10 rows, show row 0, show individual fields of row 0

d/ filter Dataset with « id > 100 and text contains ’12’ »

Answer Exercise 12

```
scala> val beansDs = spark.createDataset( (1 to 100000).map( i => CustomBean(i, i%2==0, s"Hello ${i}") ) )
beansDs: org.apache.spark.sql.Dataset[CustomBean] = [id: int, even: boolean ... 1 more field]

scala> beansDs.show(2)
+---+-----+-------+
| id| even|   text|
+---+-----+-------+
|  1|false|Hello 1|
|  2| true|Hello 2|
+---+-----+-------+
only showing top 2 rows


scala> beansDs.filter( $"id" > 100 && $"text".contains("12") ).show(4)
+---+-----+---------+
| id| even|     text|
+---+-----+---------+
|112| true|Hello 112|
|120| true|Hello 120|
|121|false|Hello 121|
|122| true|Hello 122|
+---+-----+---------+
only showing top 4 rows
```

Exercise 13 : convert to DataFrame (Dataframe = Dataset<Row>)

a/ What was the type of « ds » (from exercise 5) ?
b/ What is the type of « dsFilter » (from exercise 8) ?
c/ What is the type of « beanDs » (from exercise 12) ?

d/ Convert a Dataset of tuple « ds » (from exercise 5) to a spark DataFrame
e/ Convert a Dataset of custom Bean « beanDs » (from exercise 12) to a spark DataFrame

Answer Exercise 13

Type, then conversion:

```
scala> ds
res81: org.apache.spark.sql.Dataset[(Int, Boolean, String)] = [_1: int, _2: boolean ... 1 more field]

scala> val df = ds.toDF
df: org.apache.spark.sql.DataFrame = [_1: int, _2: boolean ... 1 more field]
```

After a filter() operation, Spark has already enterred the magic DataFrame world !

```
scala> dsFilter
res82: org.apache.spark.sql.Dataset[org.apache.spark.sql.Row] = [id: int, even: boolean ... 1 more field]
```

Converting CustomBean->DataFrame… same as Tuple -> DataFrame

```
scala> beansDs
res86: org.apache.spark.sql.Dataset[CustomBean] = [id: int, even: boolean ... 1 more field]

scala> val beansDF = beansDs.toDF
beansDF: org.apache.spark.sql.DataFrame = [id: int, even: boolean ... 1 more field]
```

Exercise 14 : Question

Do you understand the in-memory difference between ?

a/ in memory scala collection of Tuple: Set(Tuple (…))
b/ Dataset on scala Tuple: Dataset[(..Tuple)]
c/ Dataset on custom class Dataset[YourCustomBean]
c/ DataFrame …. Dataset[Row] … means YourCustomBean is the preferred spark one: « Row »

What do you think is the most efficient for developping type-safe code, and internally for Spark ?

Answer Exercise 14

Scala collection are only on the Spark **driver** side
… you may not be able to instanciate 500 Giga otects of these

Dataset (and DataFrame) are distributed over spark **executors**

To send data over the network, they are serialized then un-serialized.
So it is still real « Tuple » or « CustomBean » object pointers, but on spark executors.

Dataset[Row] is special … Spark internally is able to handle native memory / swap storage.

Exercise 15: question

When do you use { ._1, ._2 } ?

When do you use { .id, .text } ?

When do you use { .getAs[Int](« id »), .getAs[Int](« text ») } ?

Answer Exercise 15

On scala Tuple => you use _1, _2, … field names

On your custom bean => you use your class fieldName (or getter)

Example: case class YourCustomBean(id: Int, text: String, foo: String) {}

val bean = YourCustomBean(123, « text », « foo »);

bean.foo

In SQL expression, you use the column name, example « _1 », « id », « text »

On spark.sql.Row => you use org.apache.spark.sql.Row methods !!

Example: row.getAs[Int](« foo »)

Exercise 16: filtering Dataset

Using ds.filter(…) return a new dataset (dataset are immutable)
So use « ds.filter (…) .show(5, false) »

Study the different filter() methods signature available

Filter in 4 different ways with the condition

_1 >= 10 && _2 == true && _3 contains « hello »

Way 1/ using SQL where clause, 2/ using function callback
3/ using algebra on column ds.col(« _1 ») 4/ using $« _1 »

Answer Exercise 16

def filter(conditionExpr: String): org.apache.spark.sql.Dataset[(Int, Boolean, String)]

def filter(func: org.apache.spark.api.java.function.FilterFunction[(Int, Boolean, String)]): org.apache.spark.sql.Dataset[(Int, Boolean, String)]
def filter(func: ((Int, Boolean, String)) => Boolean): org.apache.spark.sql.Dataset[(Int, Boolean, String)]

def filter(condition: org.apache.spark.sql.Column): org.apache.spark.sql.Dataset[(Int, Boolean, String)]

scala> ds.filter

Answer Exercise 16

```
ds.filter( " _3 LIKE '%very%' AND _1 >= 15 ").show(3, false)

 ds.filter( row => row._1 >= 15 && row._3.contains("very") ).show(5, false)

ds.filter( ds.col("_2").and( ds.col("_1").gt ( lit(15) )) ).show(5, false)

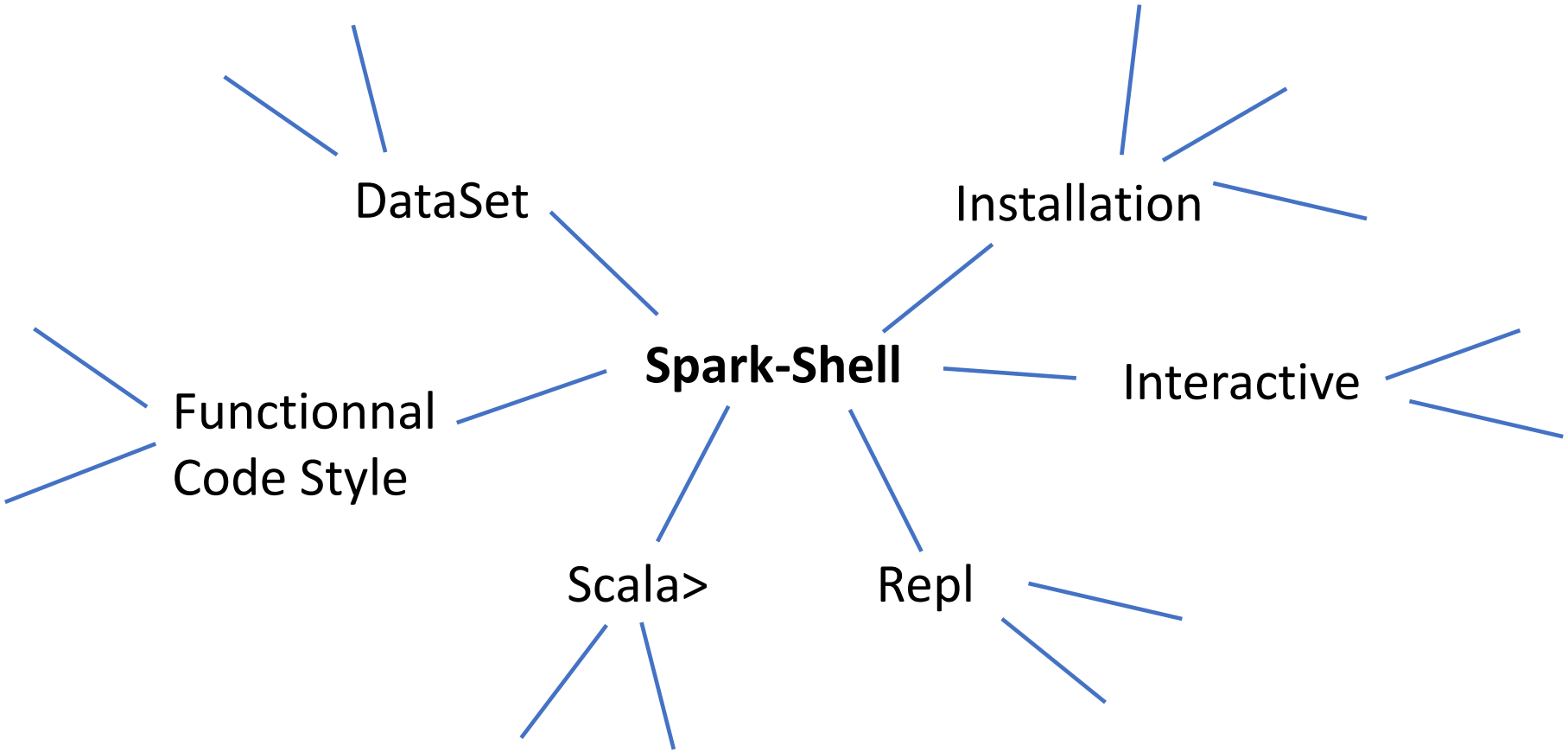
ds.filter( $"_2" .and( $"_1" .gt ( lit(15) )) ).show(5, false)
```

Exercise 17 : MindMap

Draw a MindMap to summarize
what you did and learn from this TD session

Your MindMap should
start with word « spark-shell » in the middle
Then draw star edges to other word chapters and sub-chapters

Hint Exercise 17



Objectives



1/ Install on your (Windows) PC a minimalist local SPARK



2/ Configure it, launch spark-shell



3/ Discover spark-shell scala> REPL



4/ Execute basic spark commands on DataSets

Take-Away

What You learned ?

Take Away

```
val ds = spark.createDataset( ( 1 to 100000).map ( i => ( i, 0==i%2, s"Hello ${i} ") ) )

ds.printSchema()

ds.show(10, false)

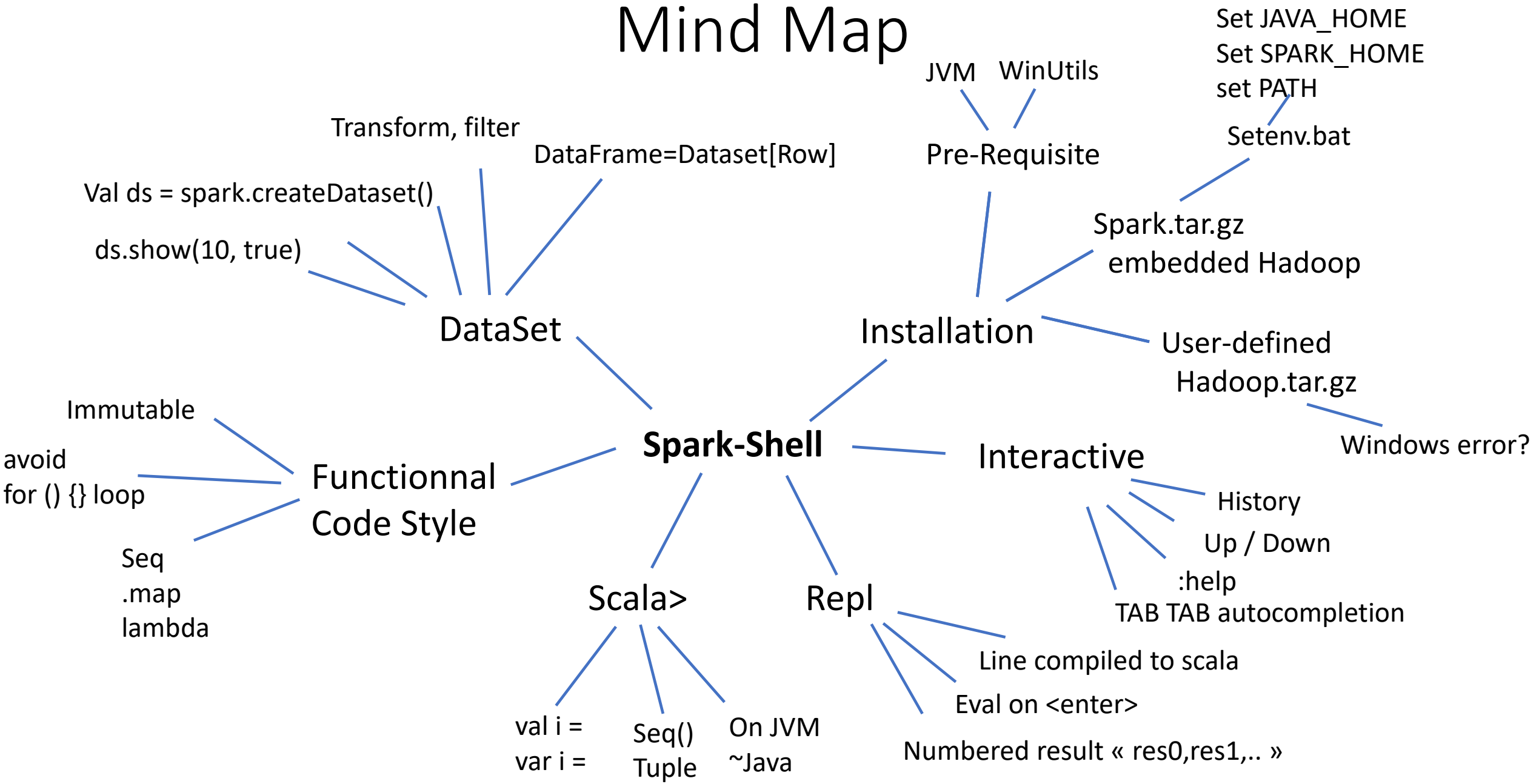
ds.filter( " _3 LIKE '%very%' AND _1 >= 15 ").show(5, false)

ds.filter( row => row._1 >= 15 && row._3.contains("very") ).show(5, false)

ds.filter( ds.col("_2").and( ds.col("_1").gt ( lit(15) )) ).show(5, false)

ds.filter( $"_2" .and( $"_1" .gt ( lit(15) )) ).show(5, false)
```

Mind Map


Spark-Shell
DataSet
Transform, filter
DataFrame=Dataset[Row]
Val ds = spark.createDataset()
ds.show(10, true)
Functionnal
Code Style
Immutable
avoid
for () {} loop
Seq
.map
lambda
Installation
Pre-Requisite
JVM
WinUtils
Spark.tar.gz
embedded Hadoop
Set JAVA_HOME
Set SPARK_HOME
set PATH
Setenv.bat
User-defined
Hadoop.tar.gz
Windows error?
Interactive
History
Up / Down
:help
TAB TAB autocompletion
Scala>
val i =
var i =
Seq()
Tuple
On JVM
~Java
Repl
Line compiled to scala
Eval on <enter>
Numbered result « res0,res1,.. »

Questions ?

Next Steps

More CMs

More TDs

Spark concepts:
- File Input / Output
- PARQUET columnar files
- SQL
- Spark Clustering
- DAG, Distribution, Optimizations
- Java binding, UDF, map
- Spark Streaming
- …